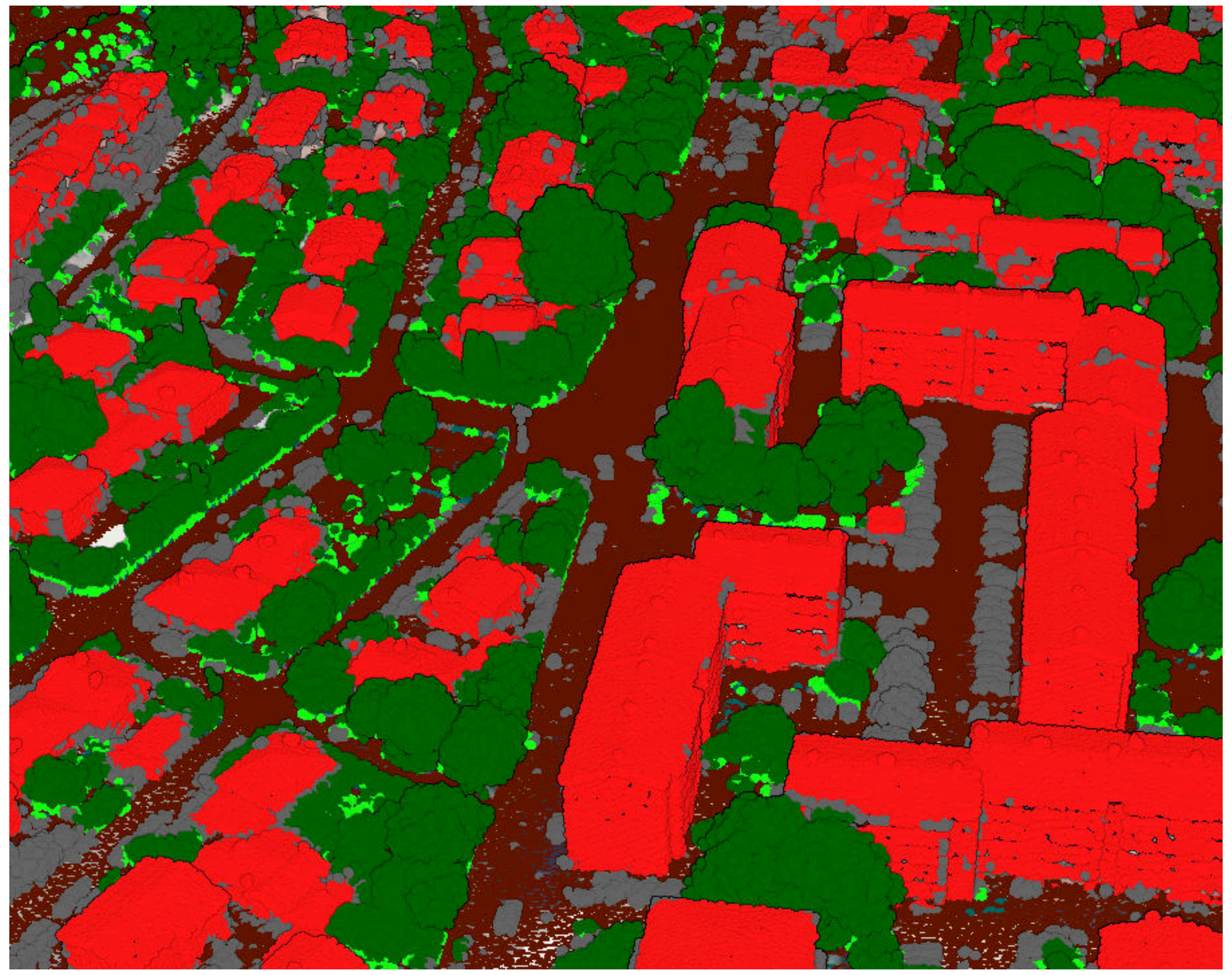
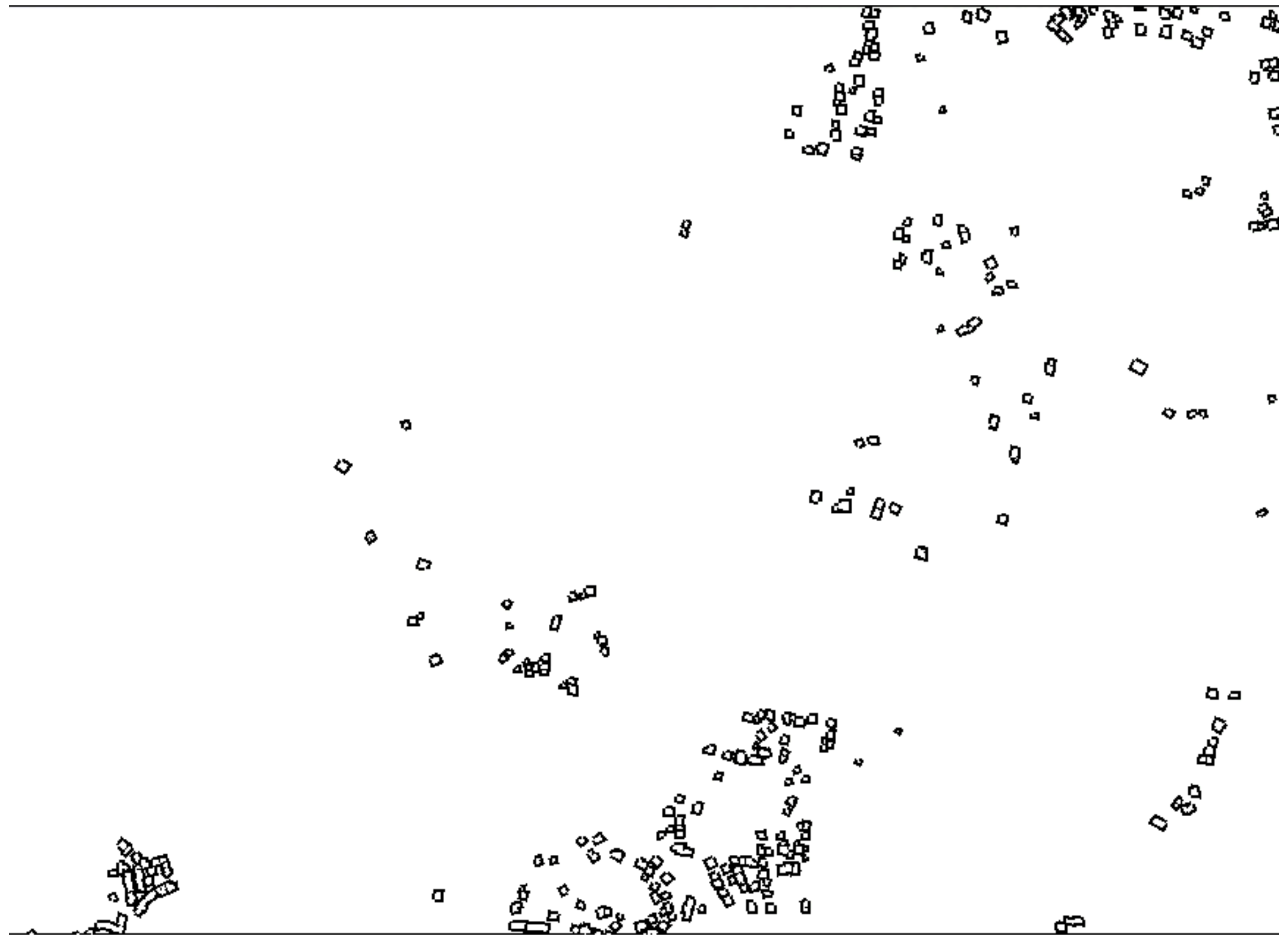
Transitions entre vues 2D et 3D pour des visualisation des données spatio-temporelles

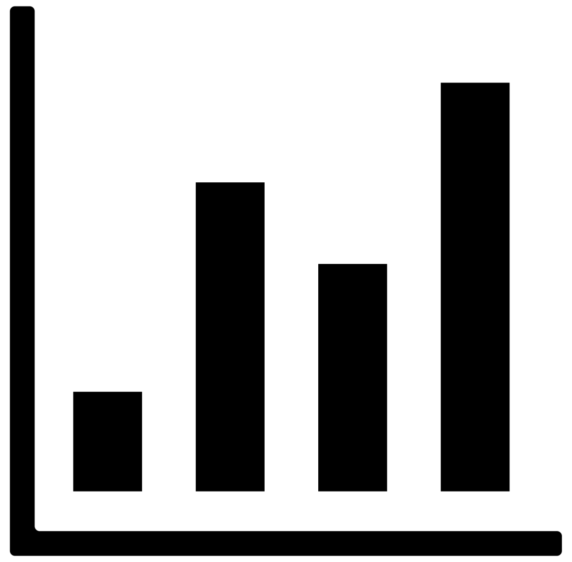
Session Visualisation de données spatio-temporelles
Journée de la recherche 2025

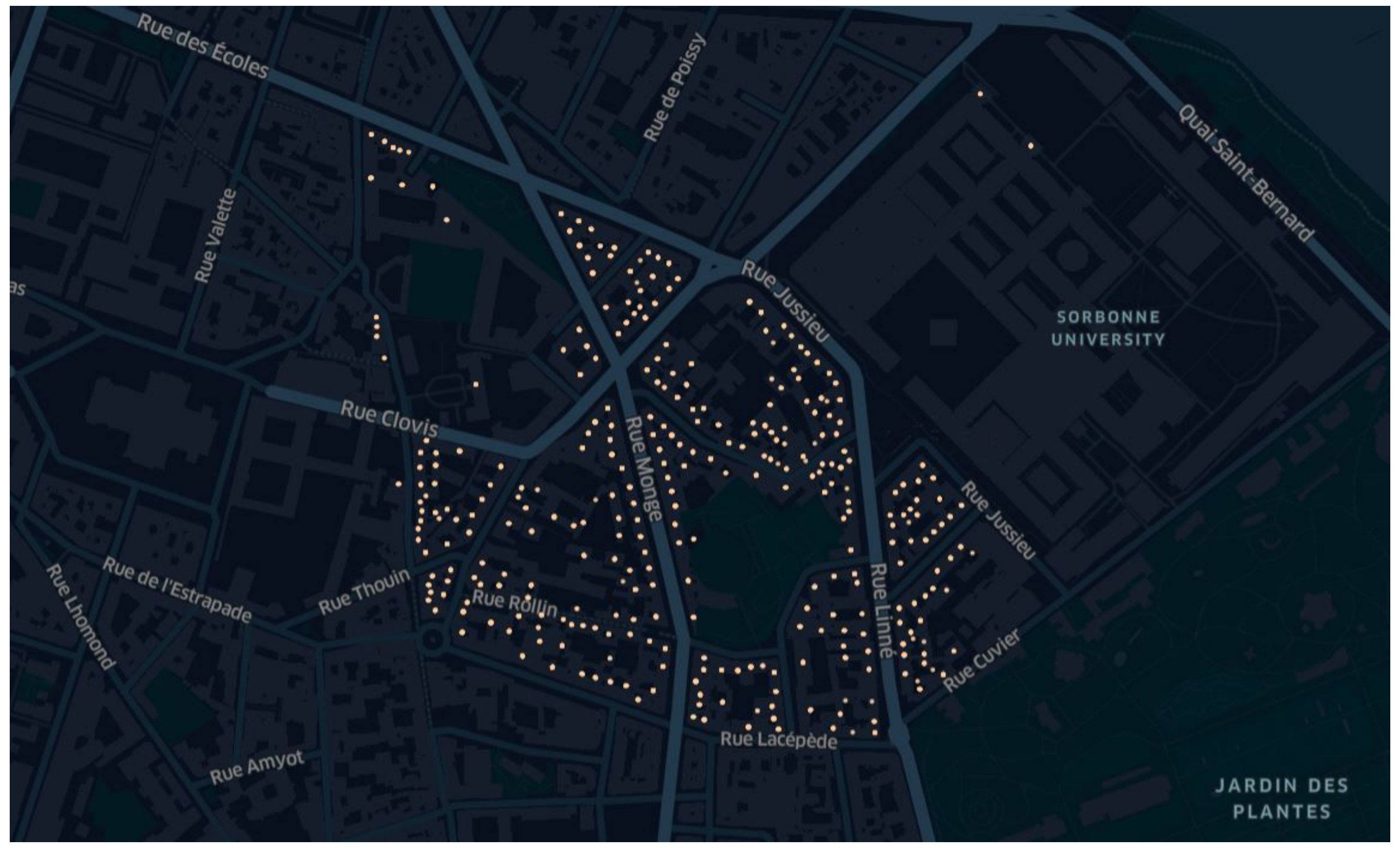
María Jesús Lobo, Jacques Gautier, Mathieu Brédif, Sidonie Christophe





INSEE

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Rue de Poissy
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Quai Saint-Bernard
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SORBONNE UNIVERSITY
Rue Clovis
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Rue de l'Estrapade
Rue Thouin
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Rue Linné
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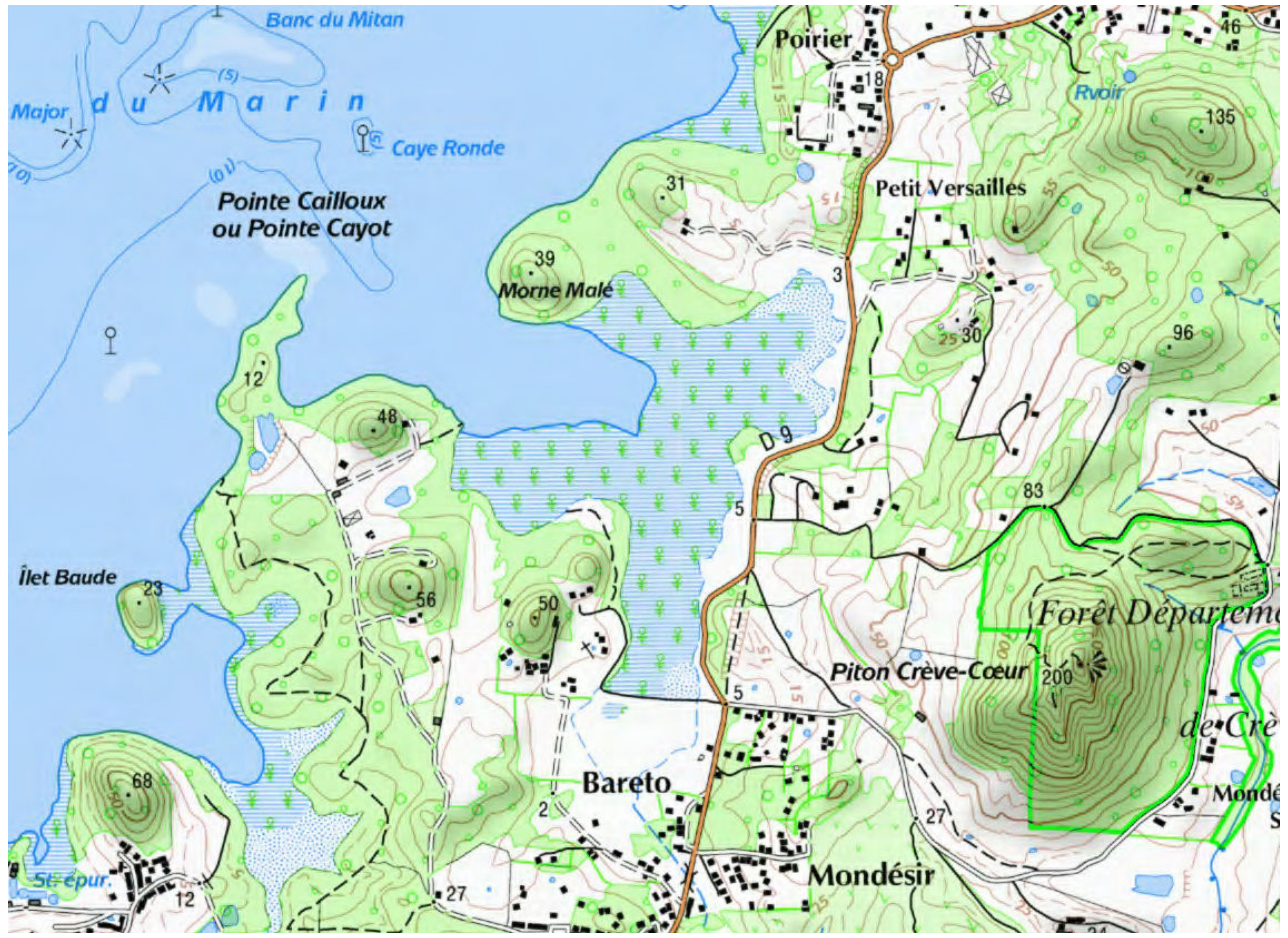
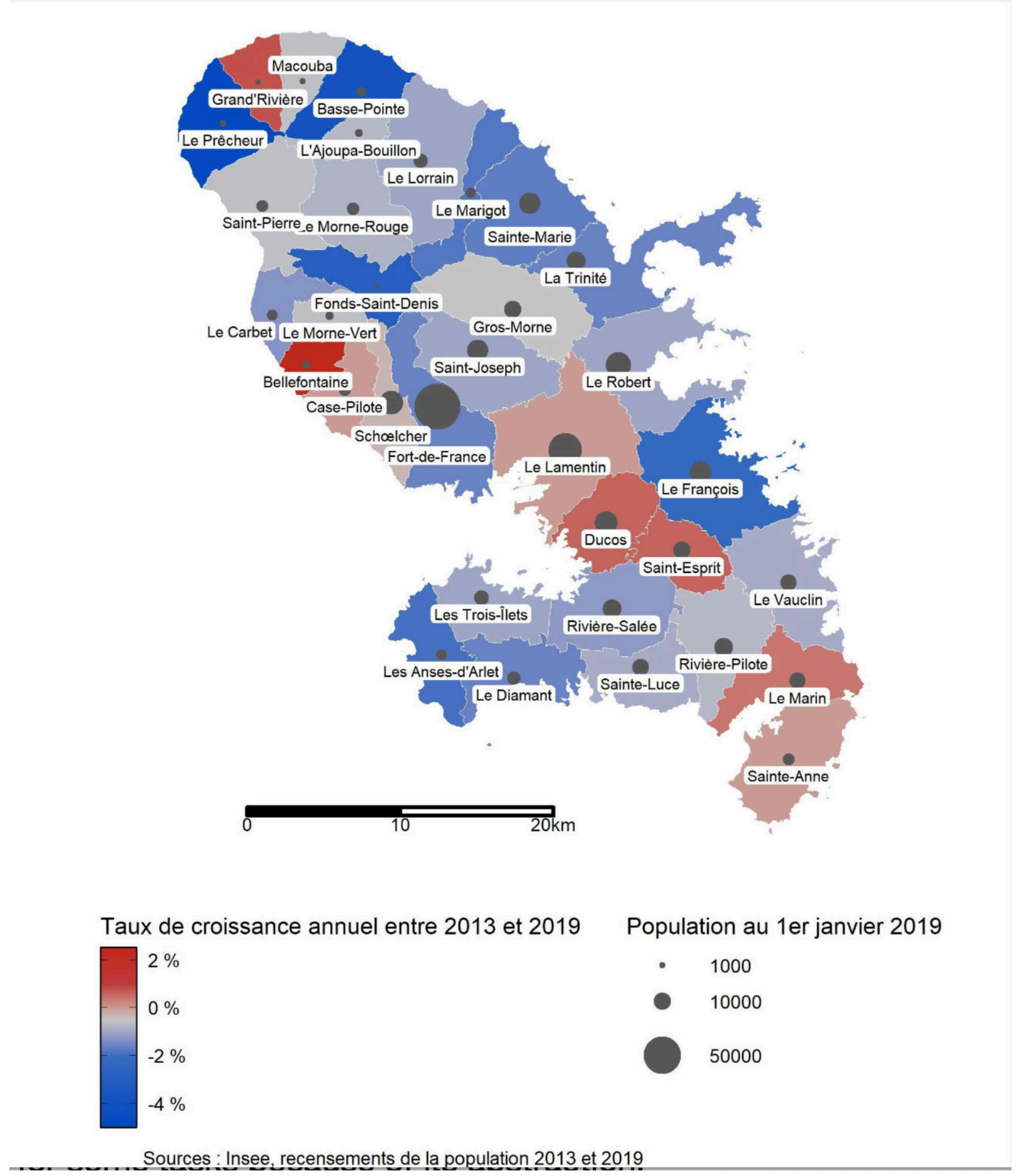
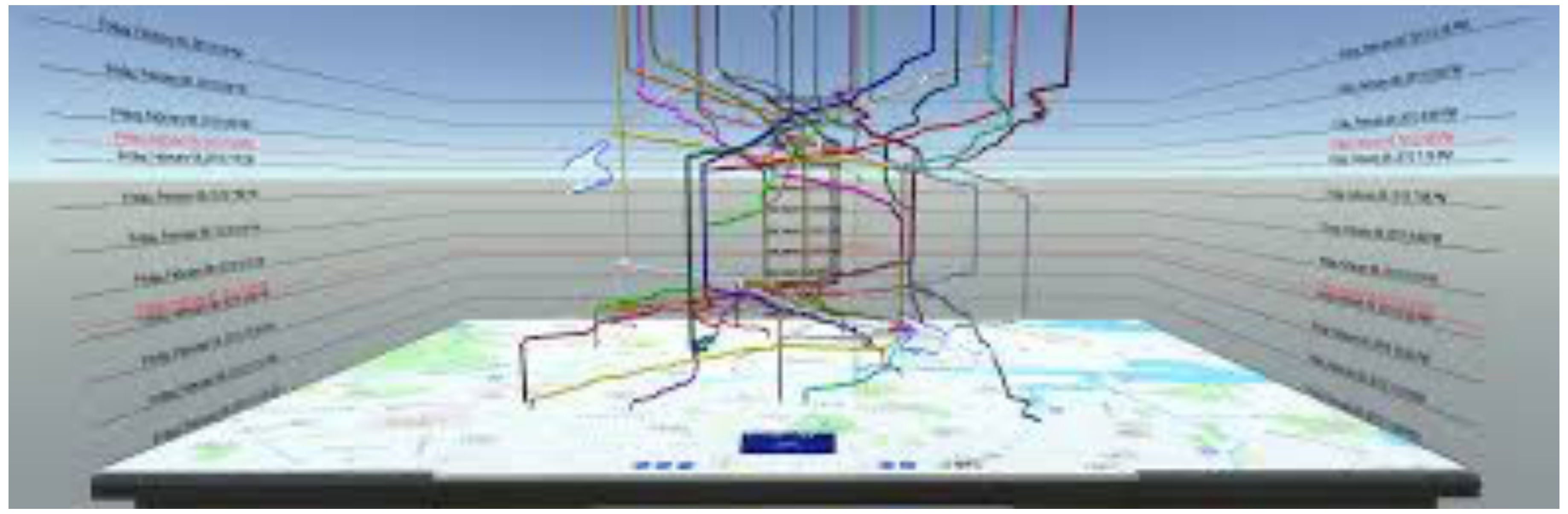
Rue Cuvier
Rue Lhomond
Rue Lacépède
Rue Amyot
JARDIN DES PLANTES



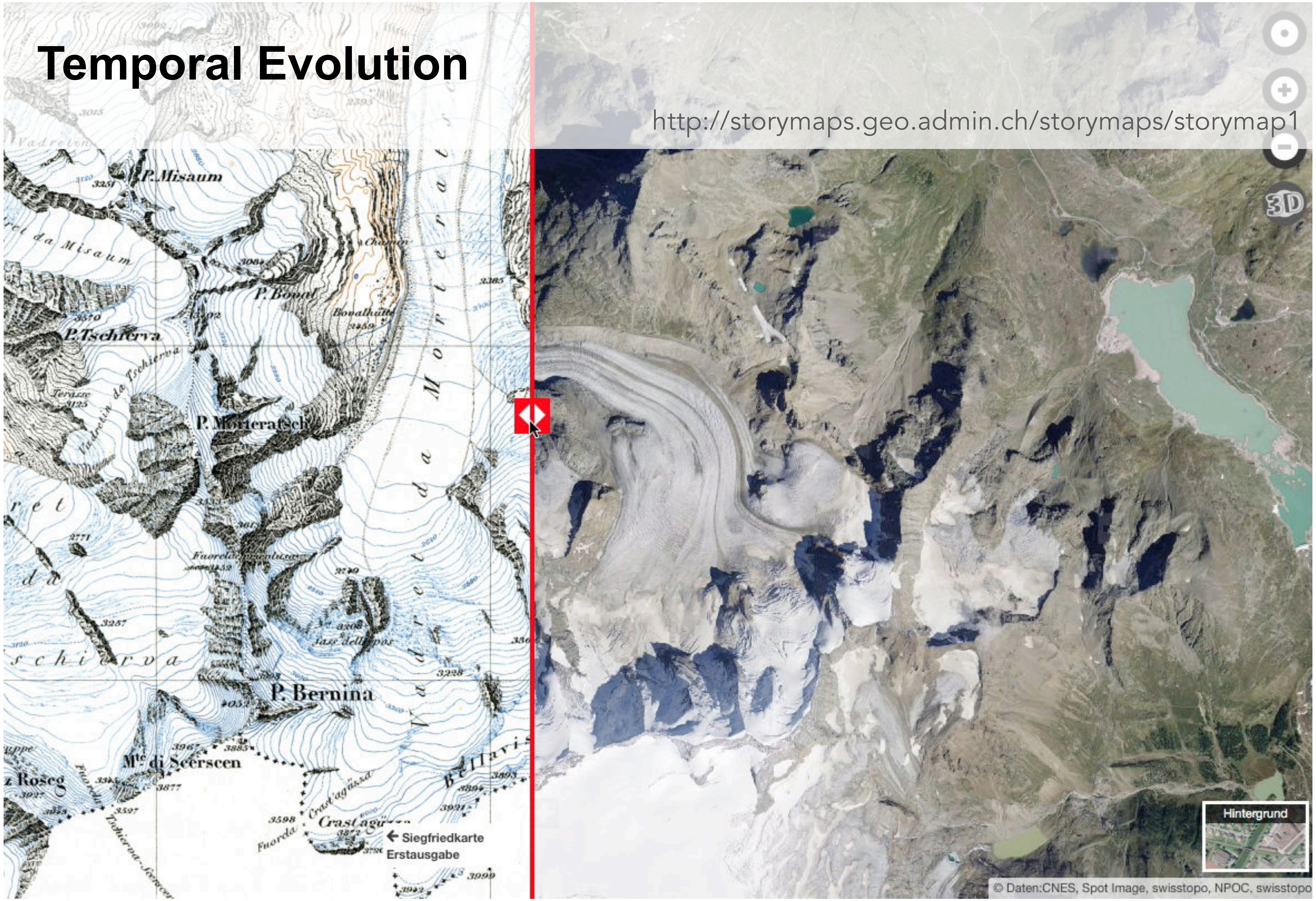




Wagner Filho, J. A., Stuerzlinger, W., & Nedel, L. (2019). Evaluating an immersive space-time cube geovisualization for intuitive trajectory data exploration. *IEEE transactions on visualization and computer graphics*, *26*(1), 514-524.

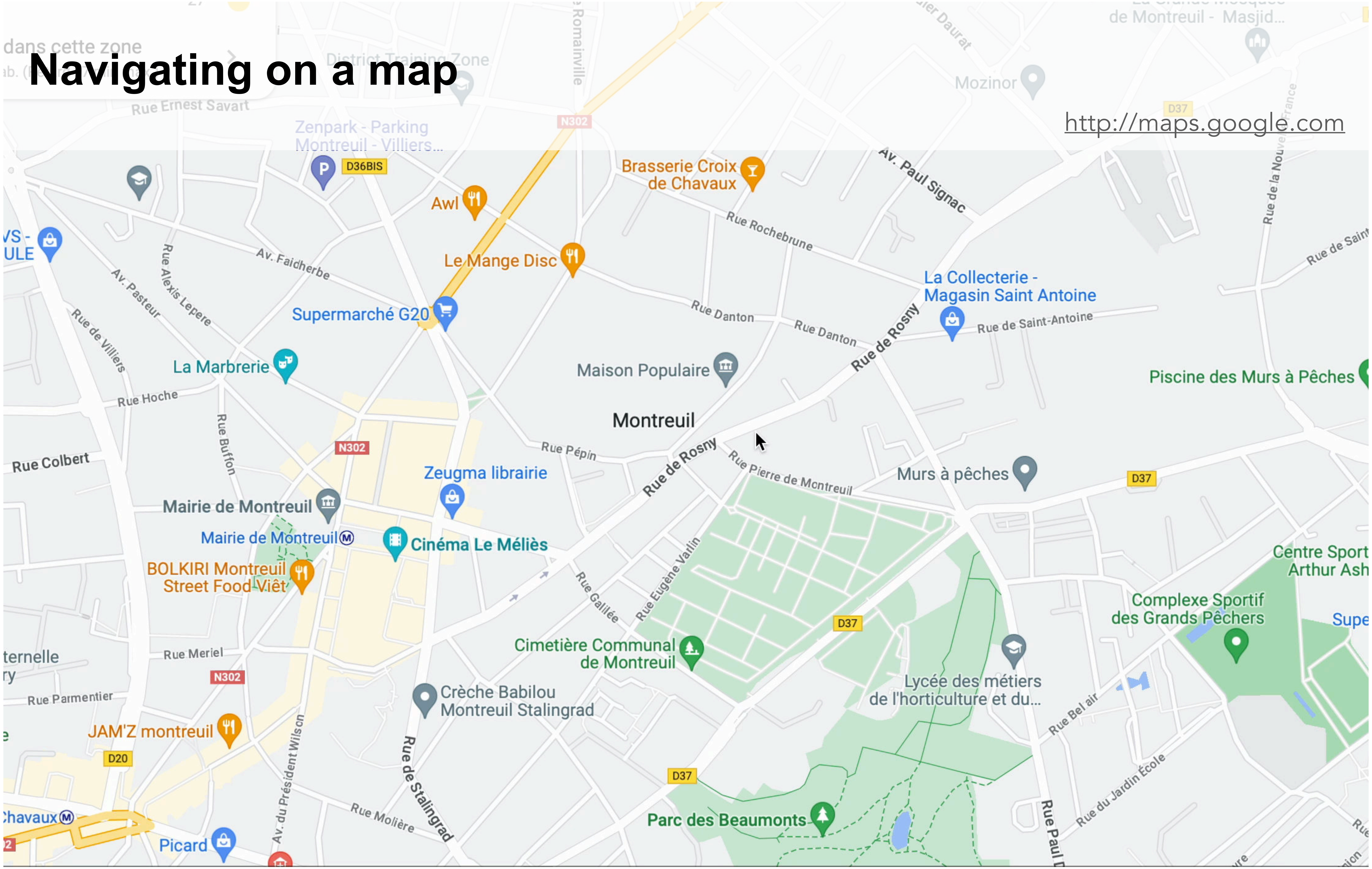
Dynamic displays and **interactivity** help users **combine multiple** geographical representations

Temporal Evolution



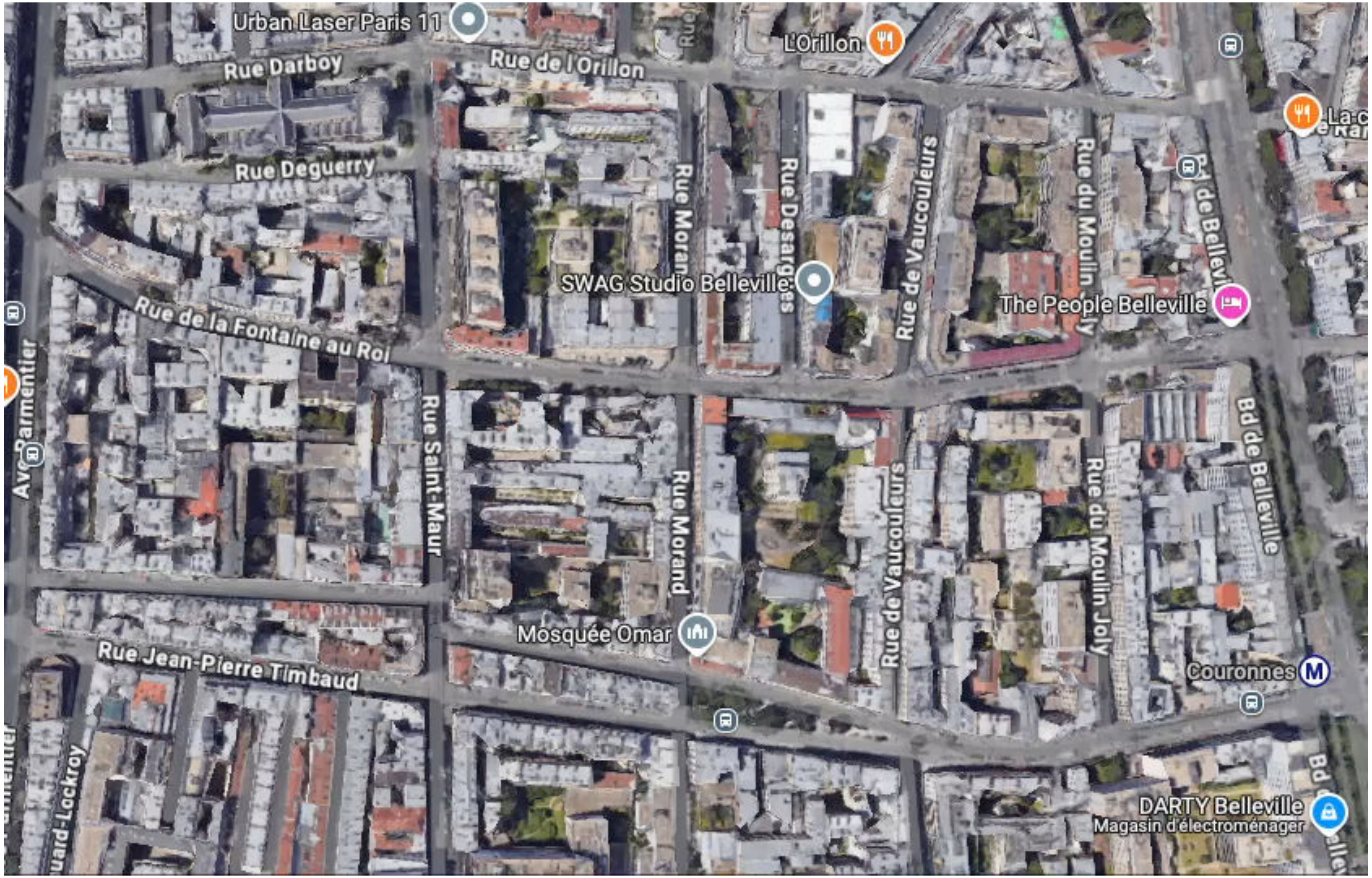
Navigating on a map

http://maps.google.com



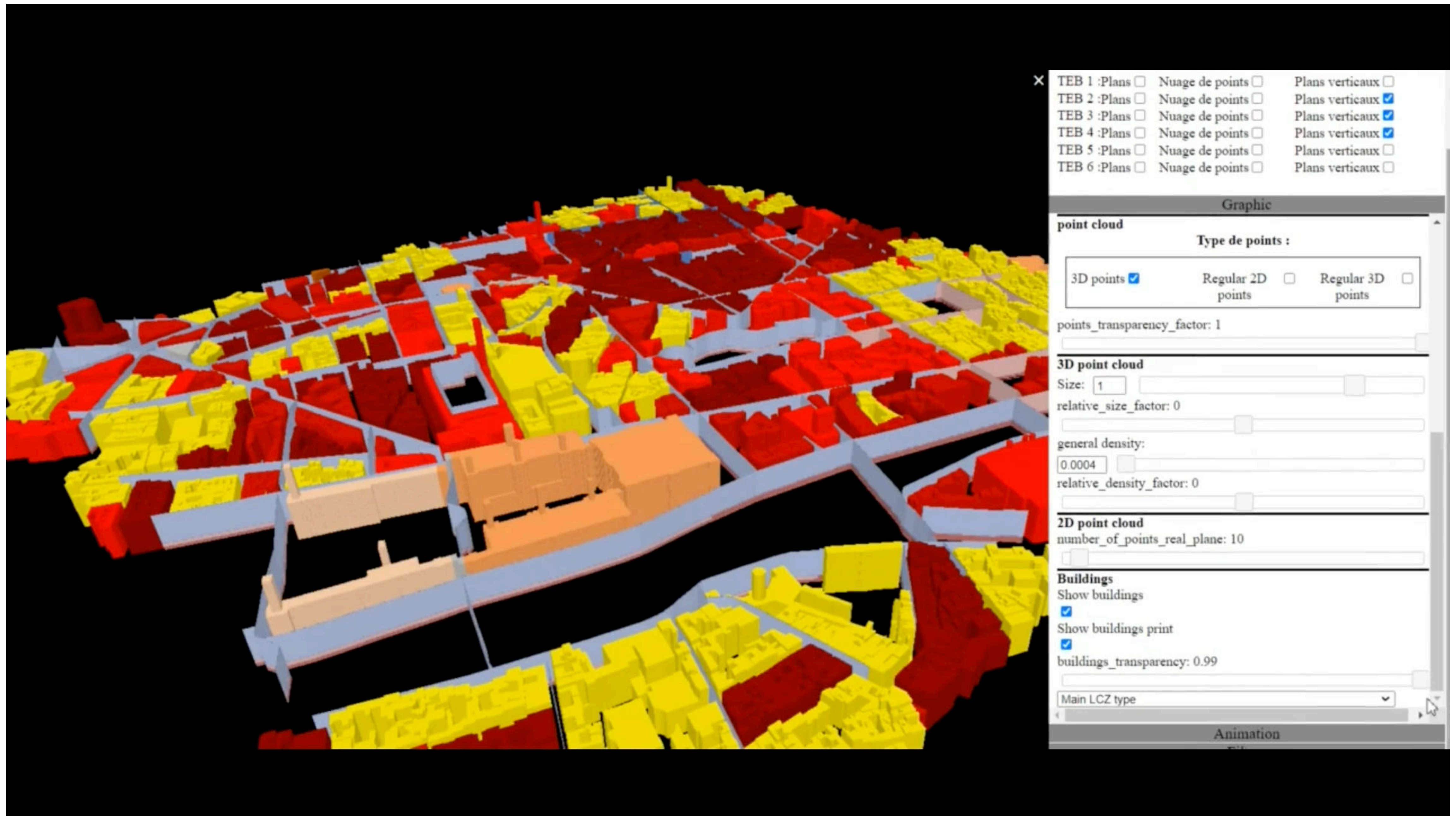
The problem is even more complicated when combining and navigating in 3D visualisations…

Navigating in a realistic 3D map

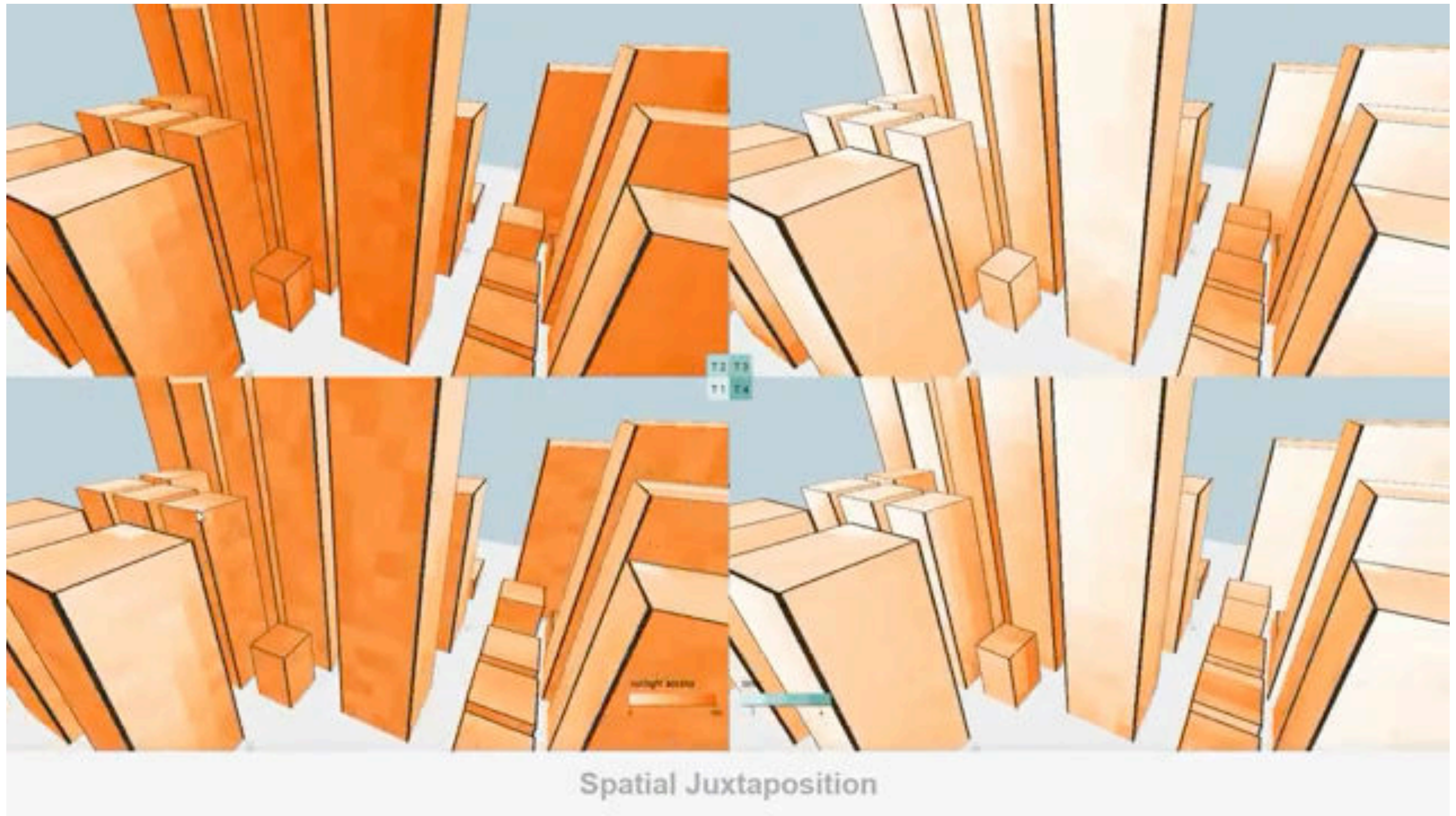


http://maps.google.com

Visualizing urban models and additional data



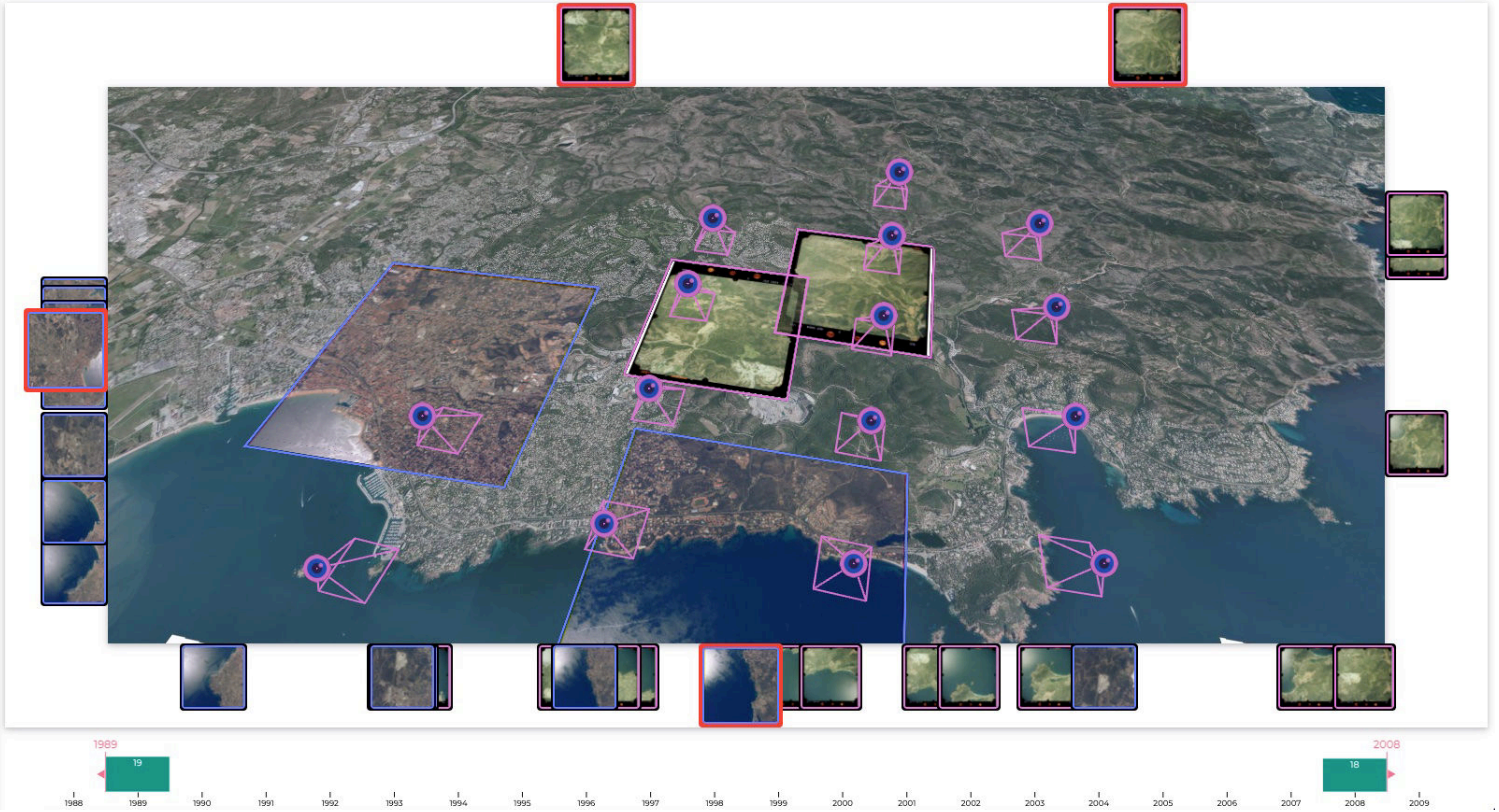
Gautier, J., Brédif, M., & Christophe, S. (2020, October). Co-visualization of air temperature and urban data for visual exploration. In 2020 IEEE Visualization Conference (VIS) (pp. 71-75). IEEE.



Mota, R., Ferreira, N., Silva, J. D., Horga, M., Lage, M., Ceferino, L., ... & Miranda, F. (2022). A comparison of spatiotemporal visualizations for 3D urban analytics. *IEEE transactions on visualization and computer graphics*, *29*(1), 1277-1287.

… or when combining 2D and 3D representations

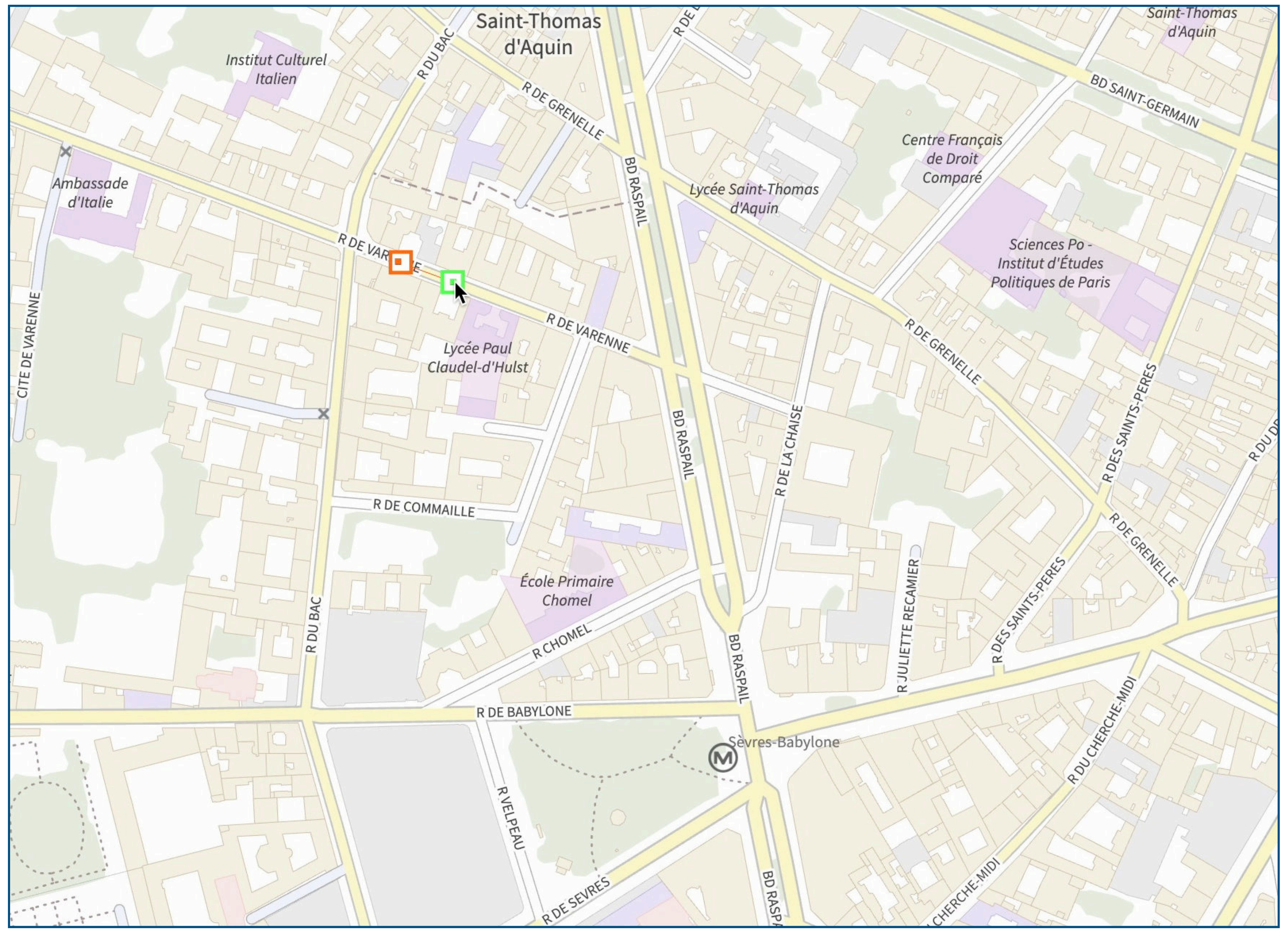
Adding 2D images in top of a 3D environment



Paiz-Reyes, Brédif, Christophe ISPRS2020, PhD thesis 2021

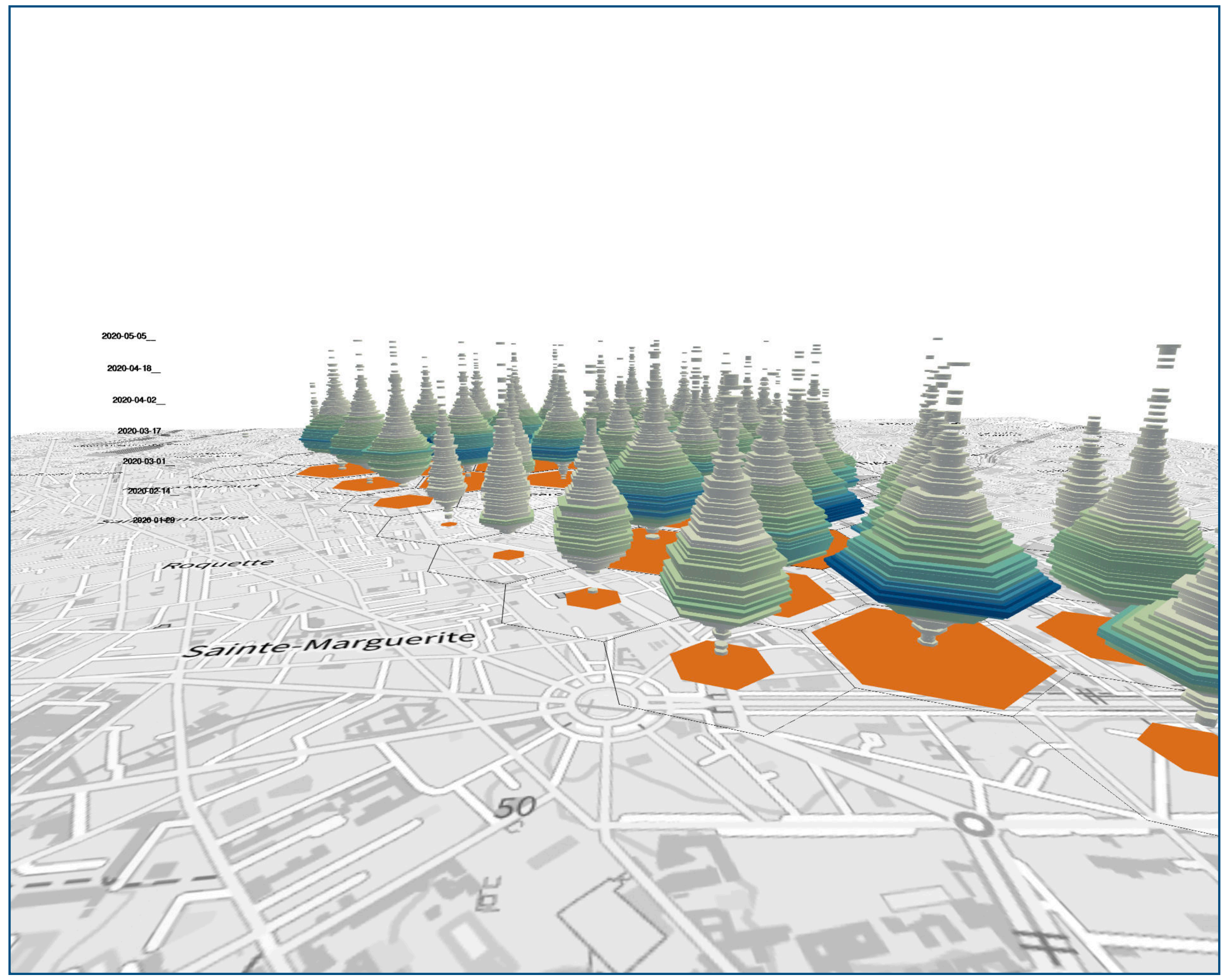
Can we propose **interactive transitions** that support users when combining **2D** and **3D** data **representations,** from different **point of views** and with **different representations**?

Map views



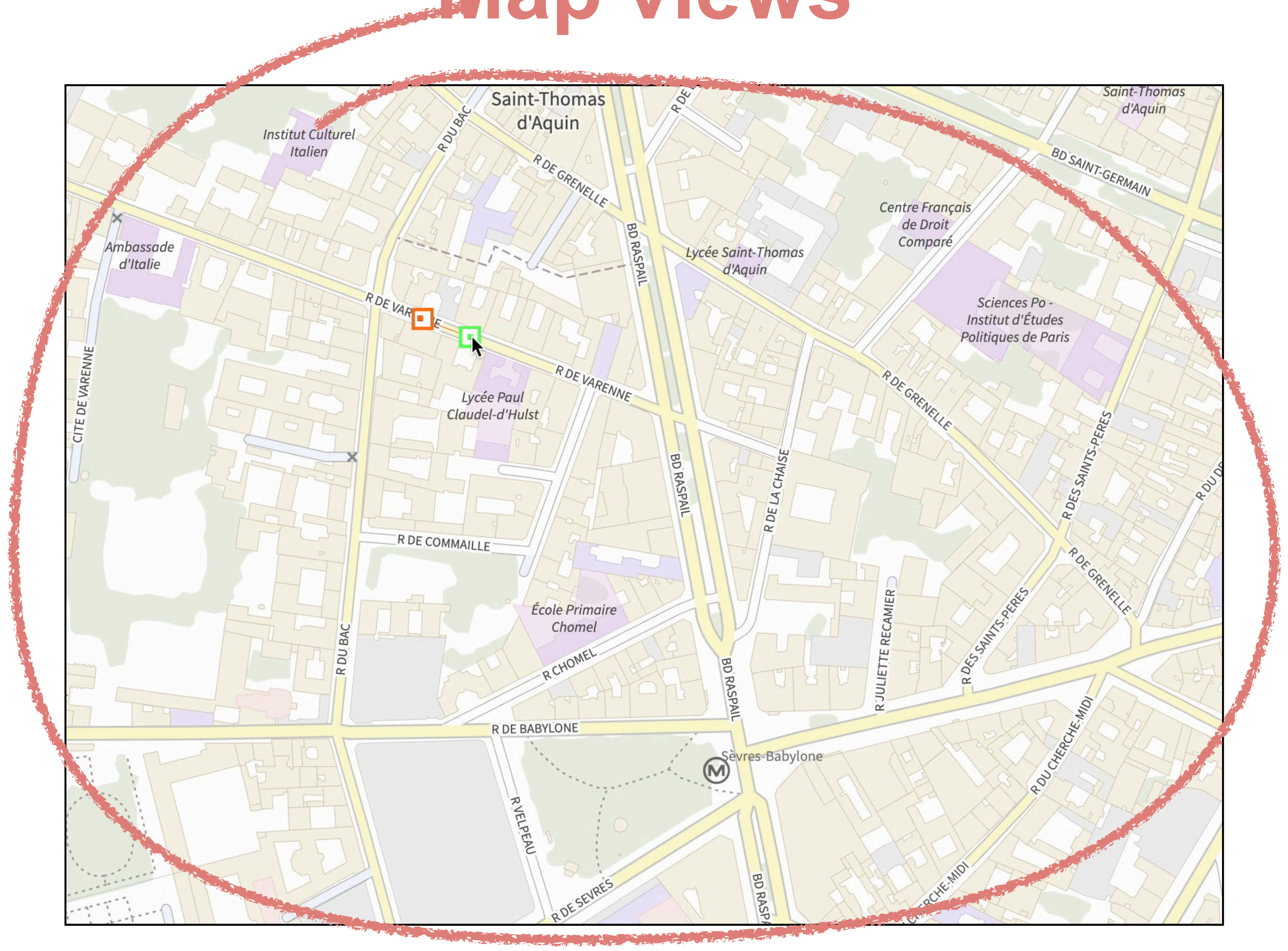
2D top-down to 3D immersive transitions

Spatio-temporal visualisations



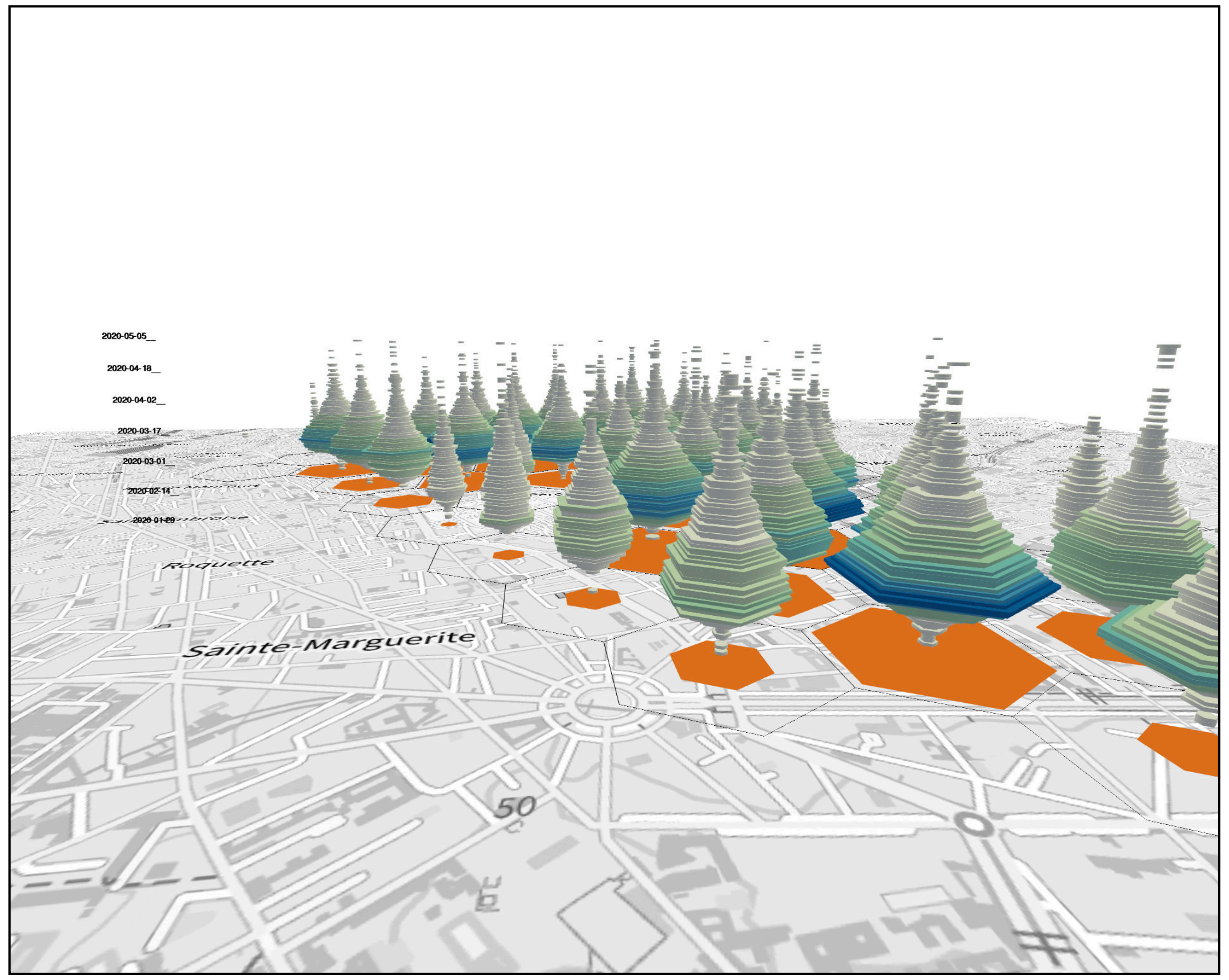
Géopidemio

Map views



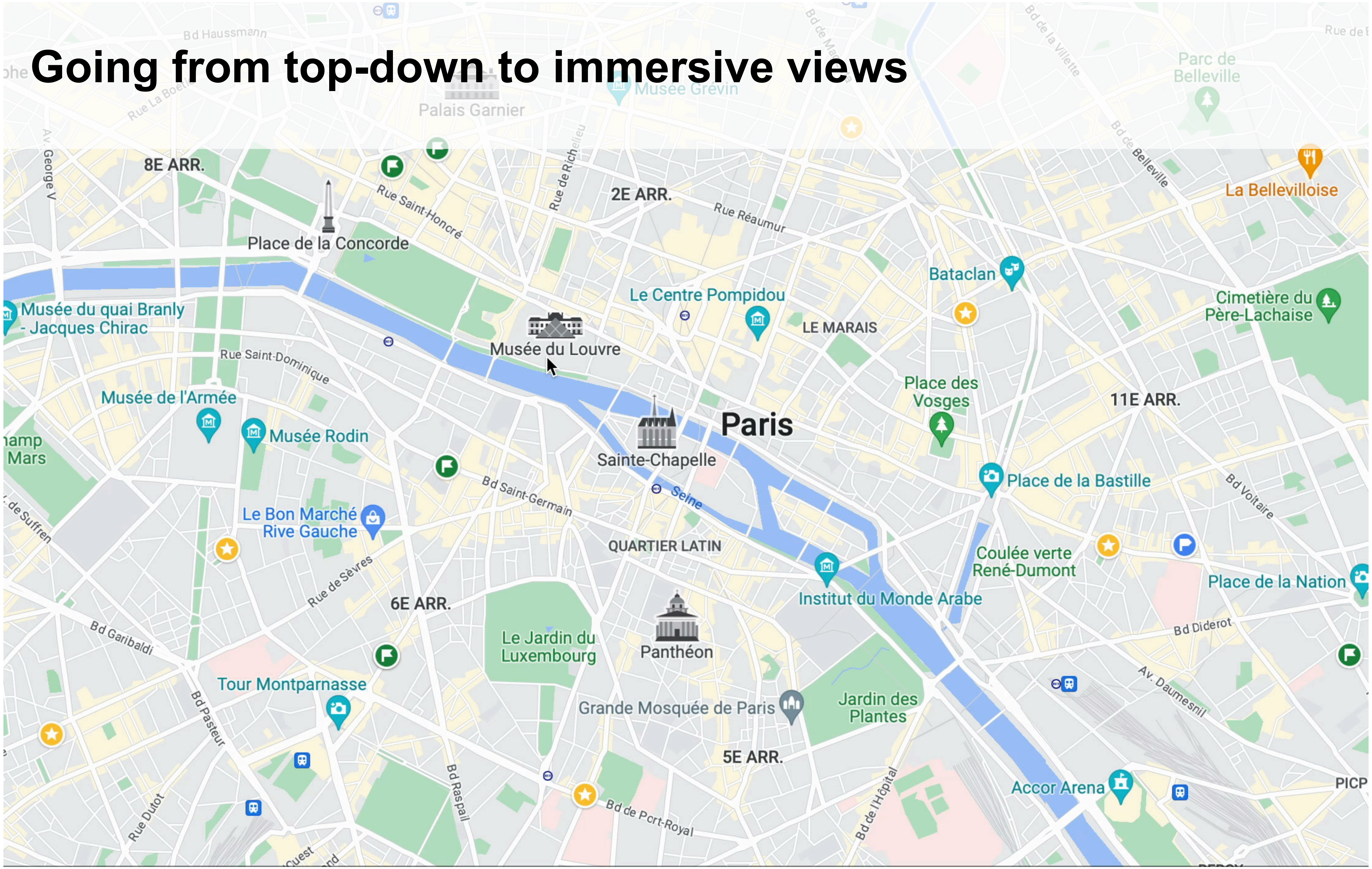
2D top-down to 3D immersive transitions

Spatio-temporal visualisations



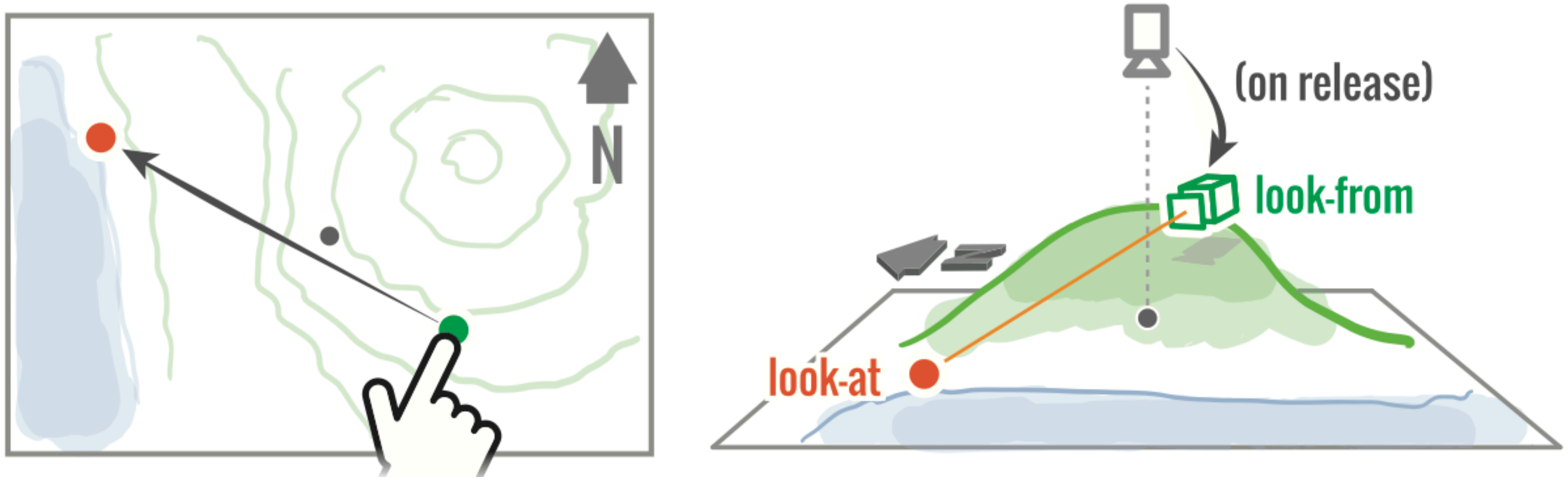
Géopidemio

Going from top-down to immersive views



This **transitions** are difficult to understand because they change **simultaneously** the **camera** parameters and the **style** of the representation.

From-at metaphor (Danyluk et al.,2019)



Danyluk, K., Jenny, B., & Willett, W. (2019, May). Look-from camera control for 3D terrain maps. In Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems (pp. 1-12).

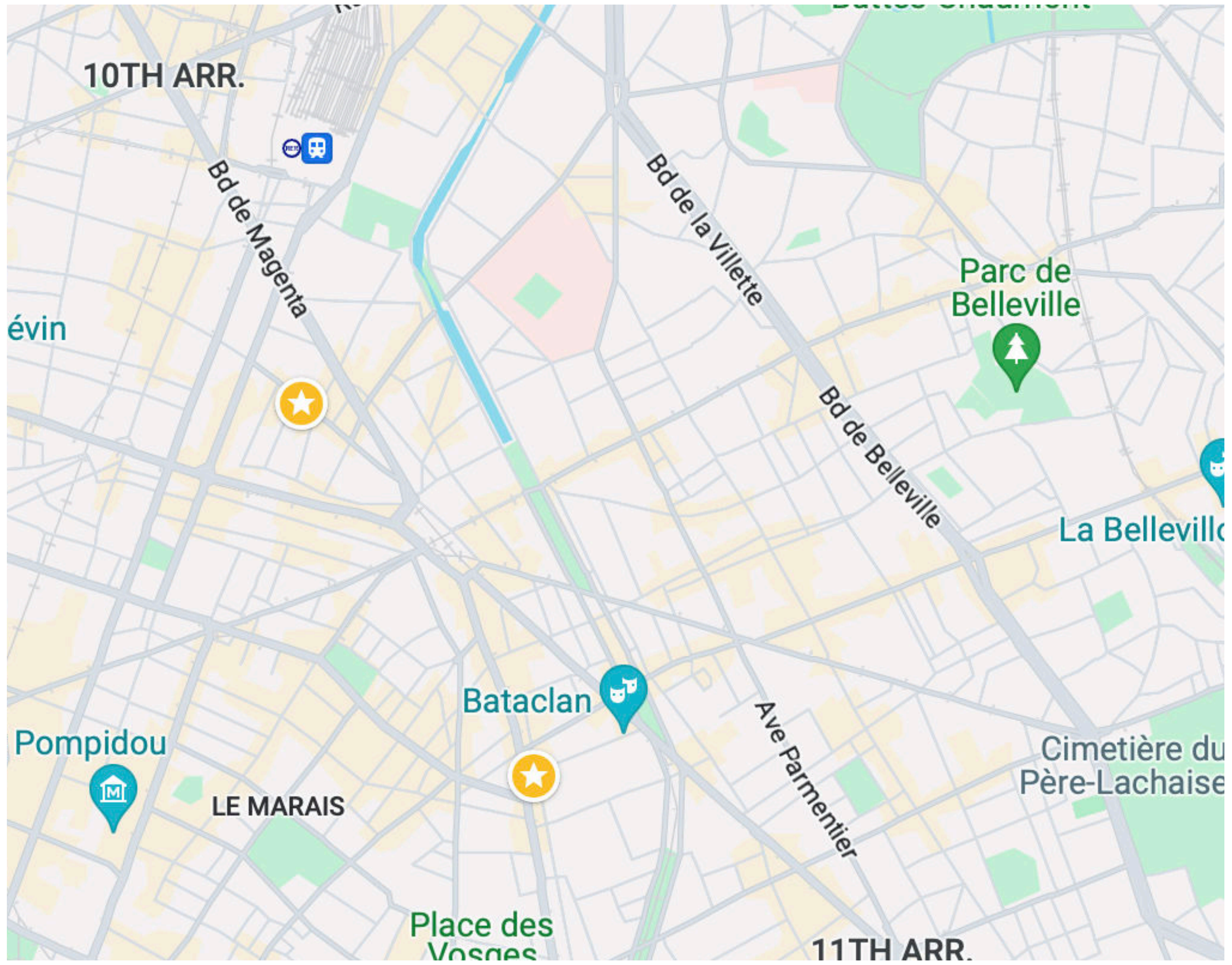
Transition requirements

According to Danyluk et al., for transitions to be effective they should:

- Keep **look-from points** in frame during transitions
- **Sequence** camera **rotation/pitch** to minimise roll

We consider separately the ***transition*** and the ***interaction technique,*** how the user defines and controls the transition

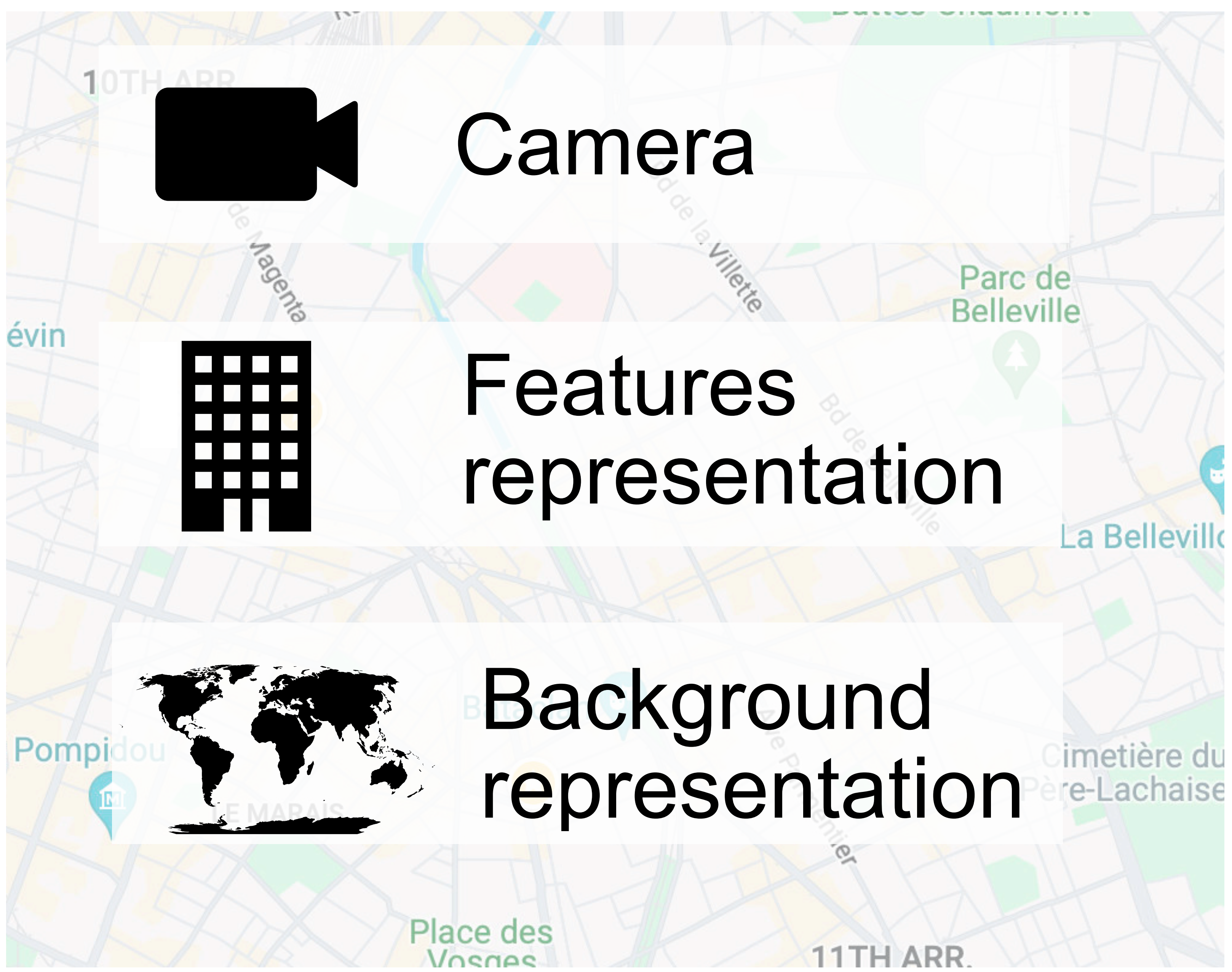
Transition: how the representation is changed



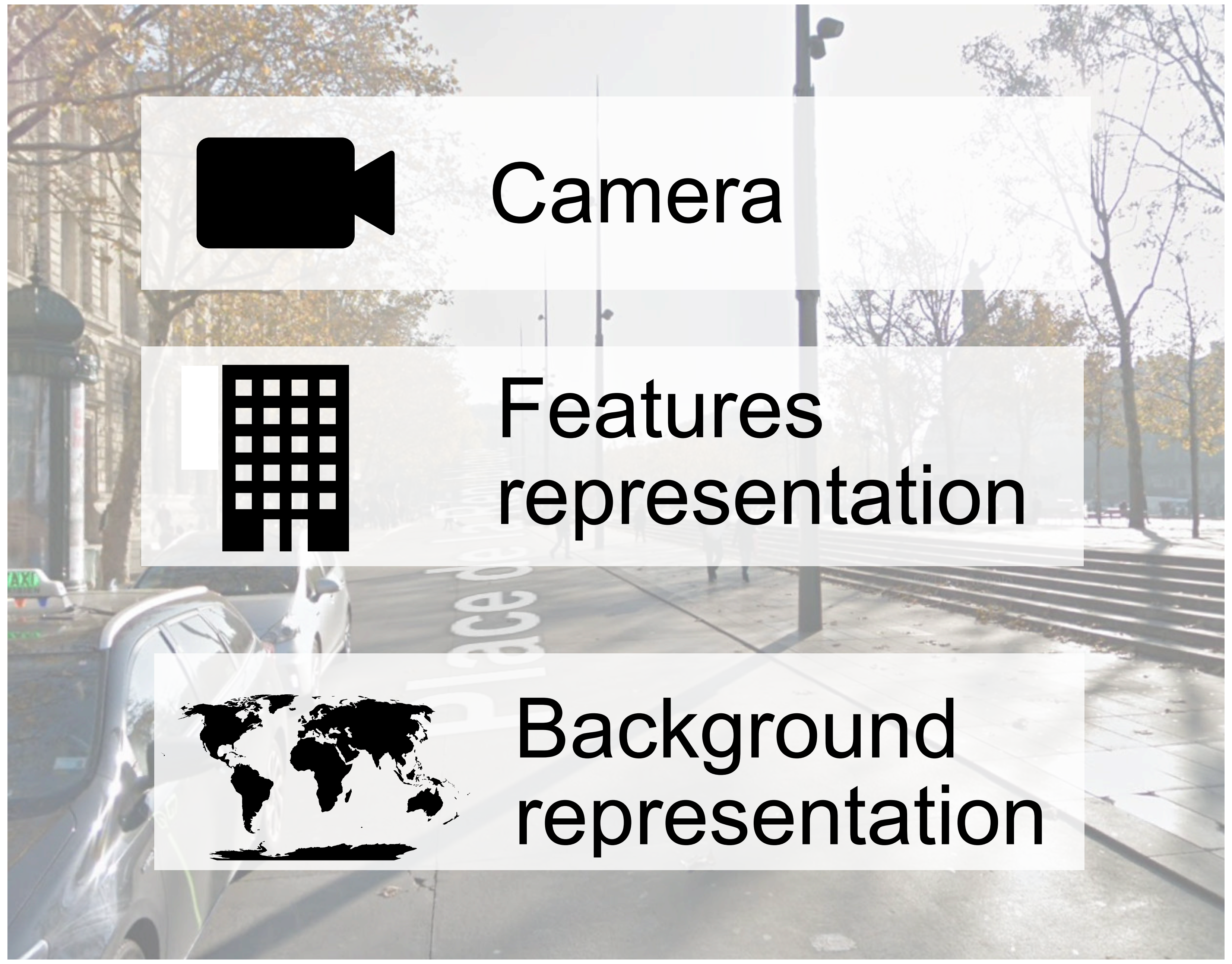
Start view

End View

Transition: how the representation is changed

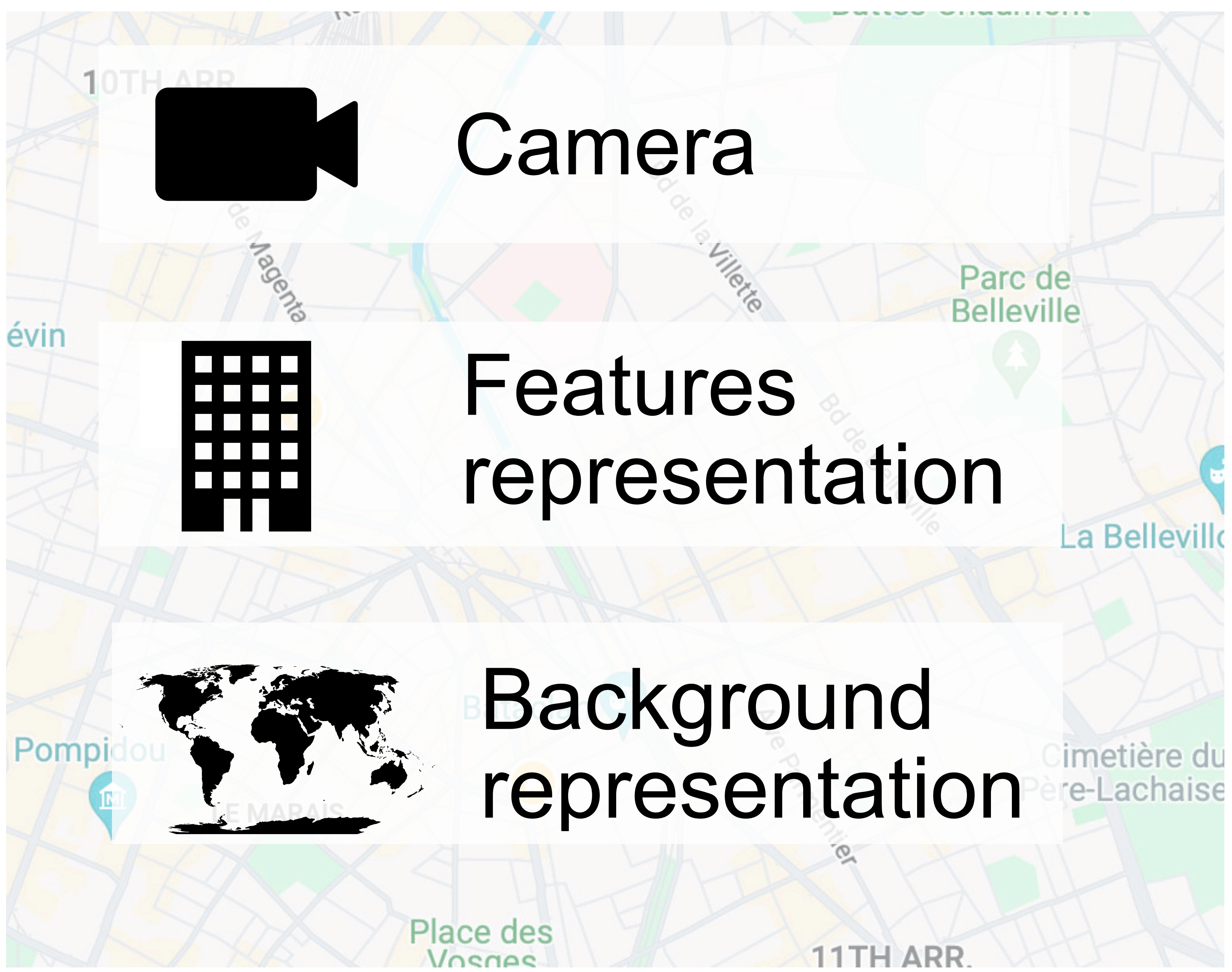


Start view



End view

Transition: how the representation is changed

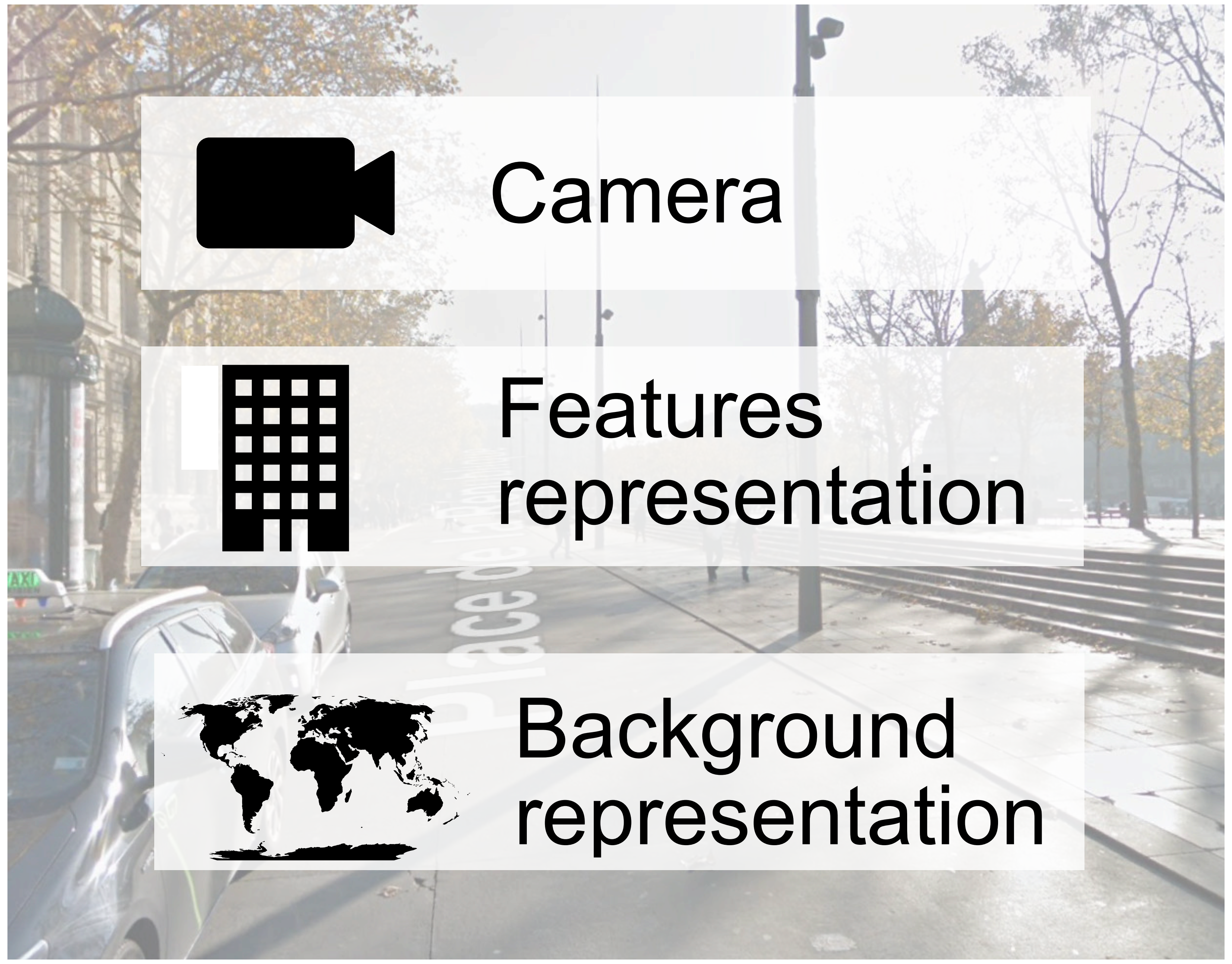


Start view

Camera transition

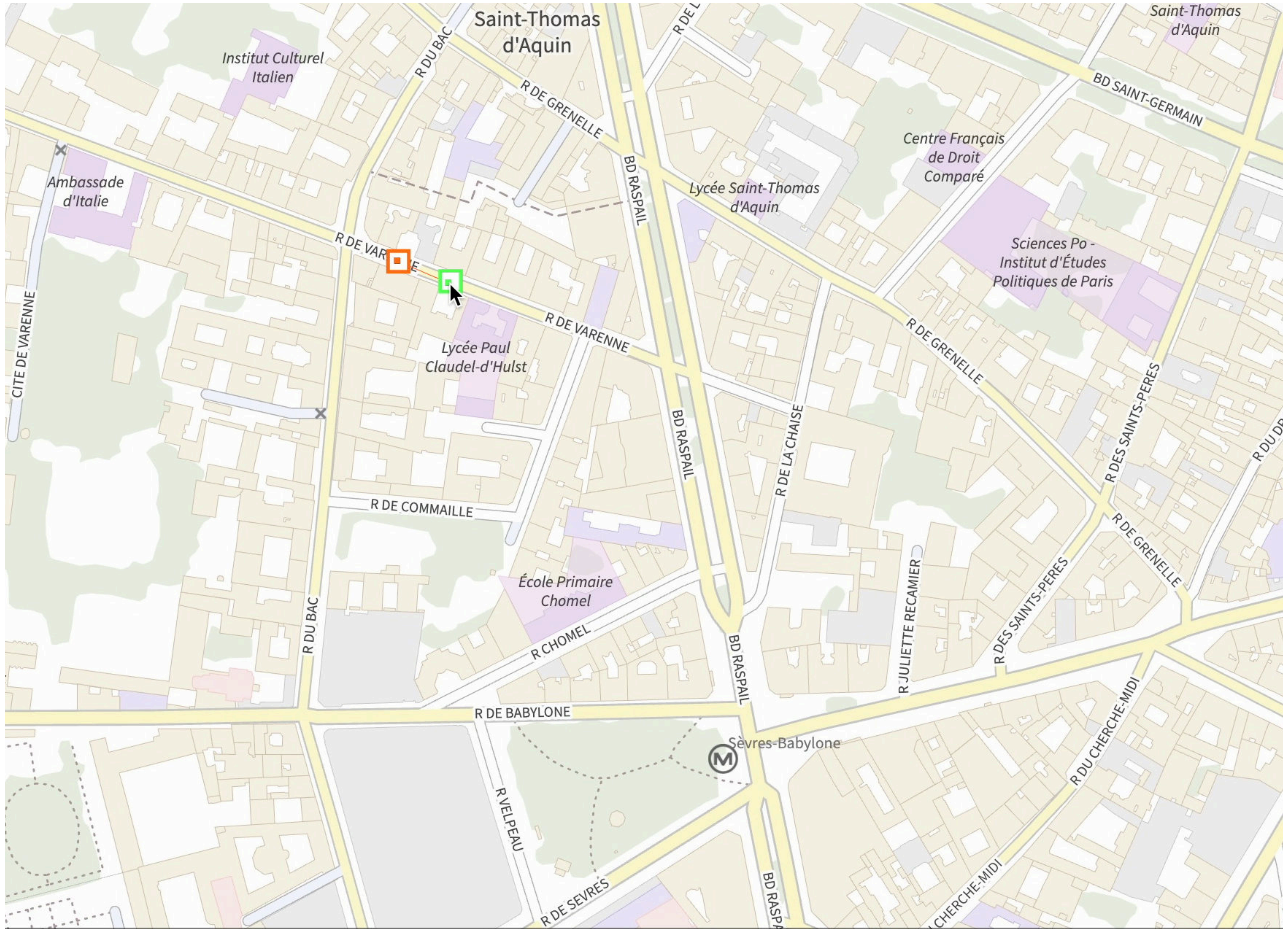
Features transition

Background Transition

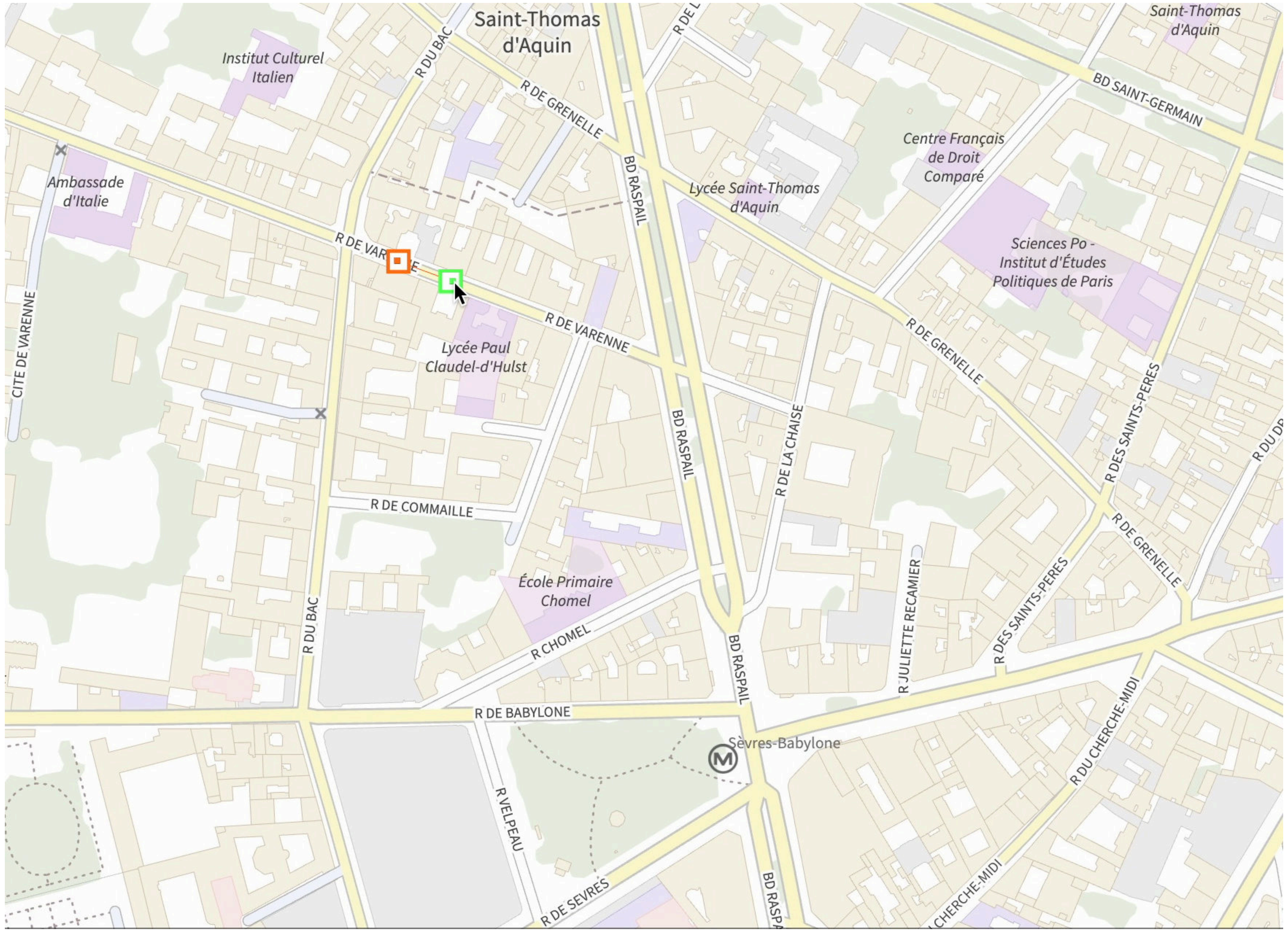


End view

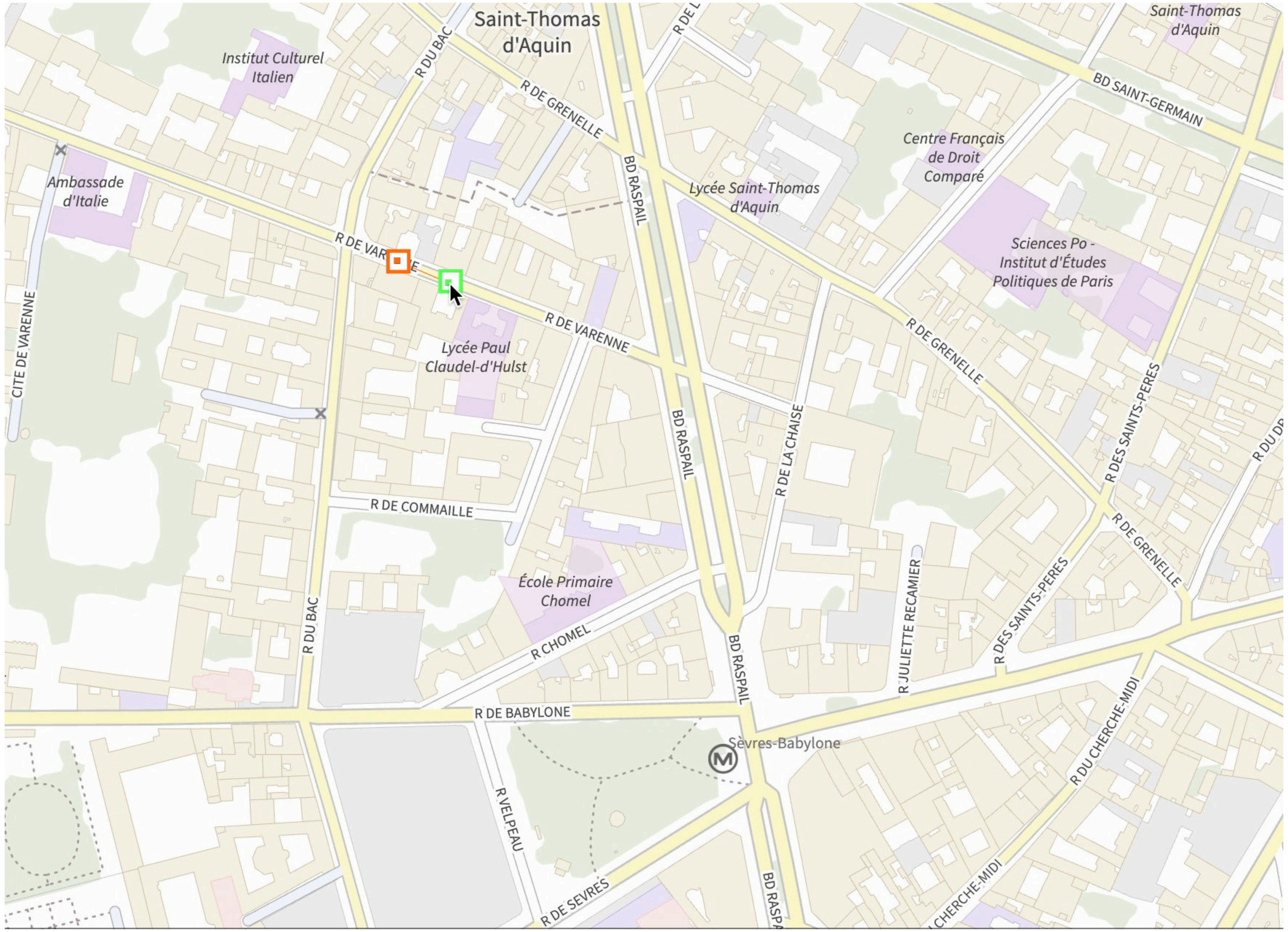
Proposed camera transitions:
Rotation First (RF)



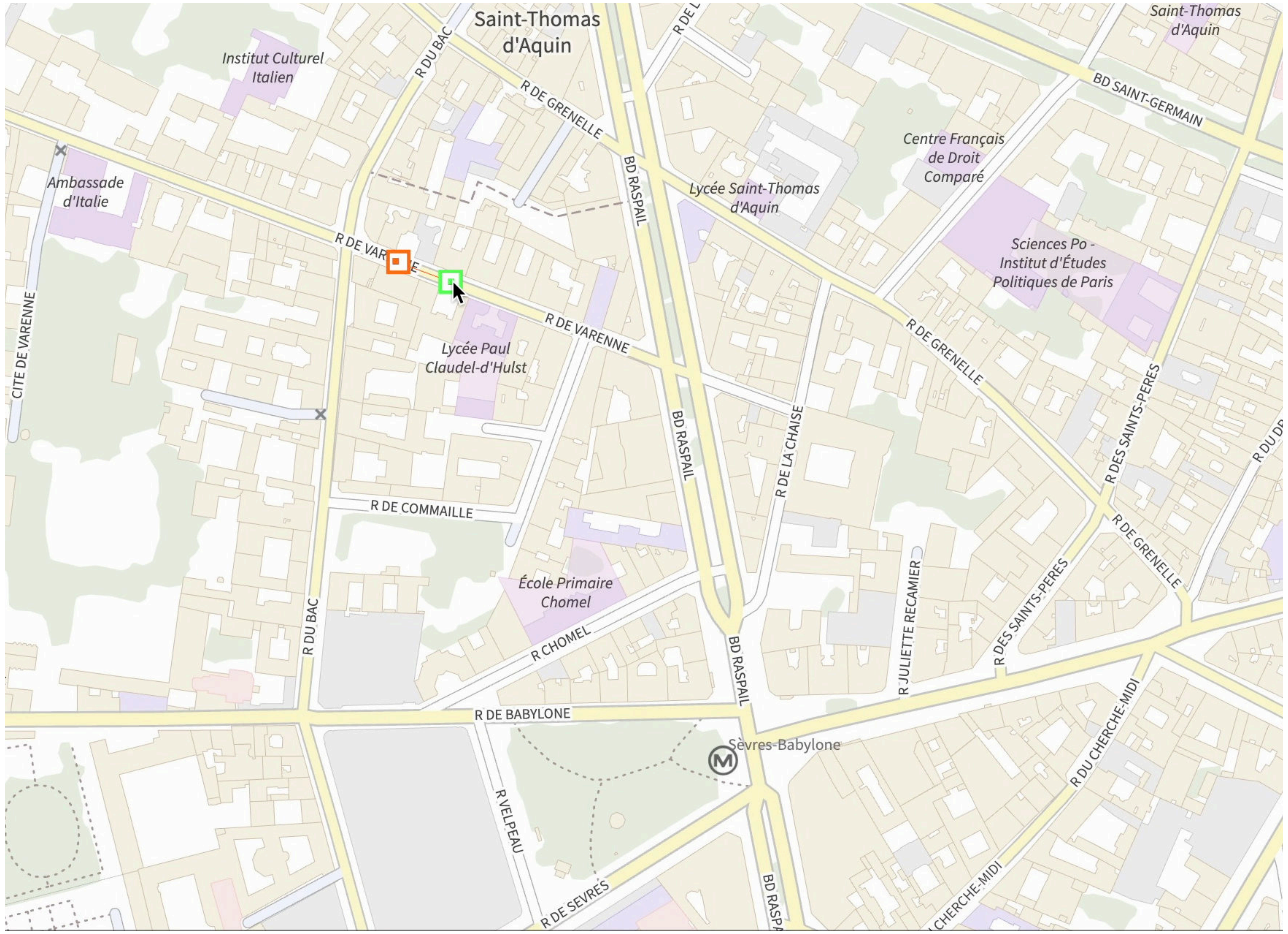
Proposed camera transitions:
Zoom First (RF)



Proposed transition orders:
Camera First (CF)

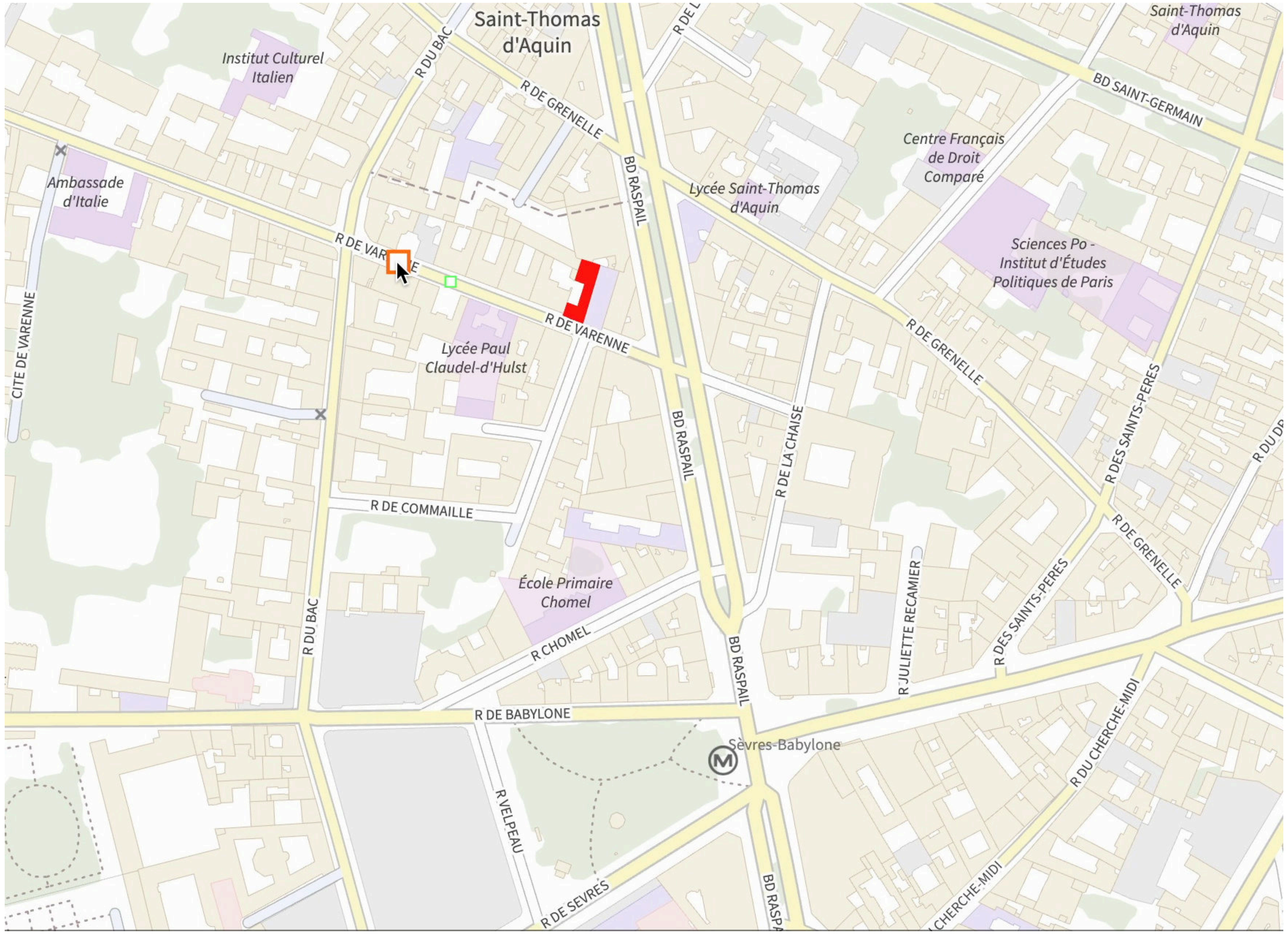


Proposed transition orders:
Features First (FF)



Proposed transition orders:

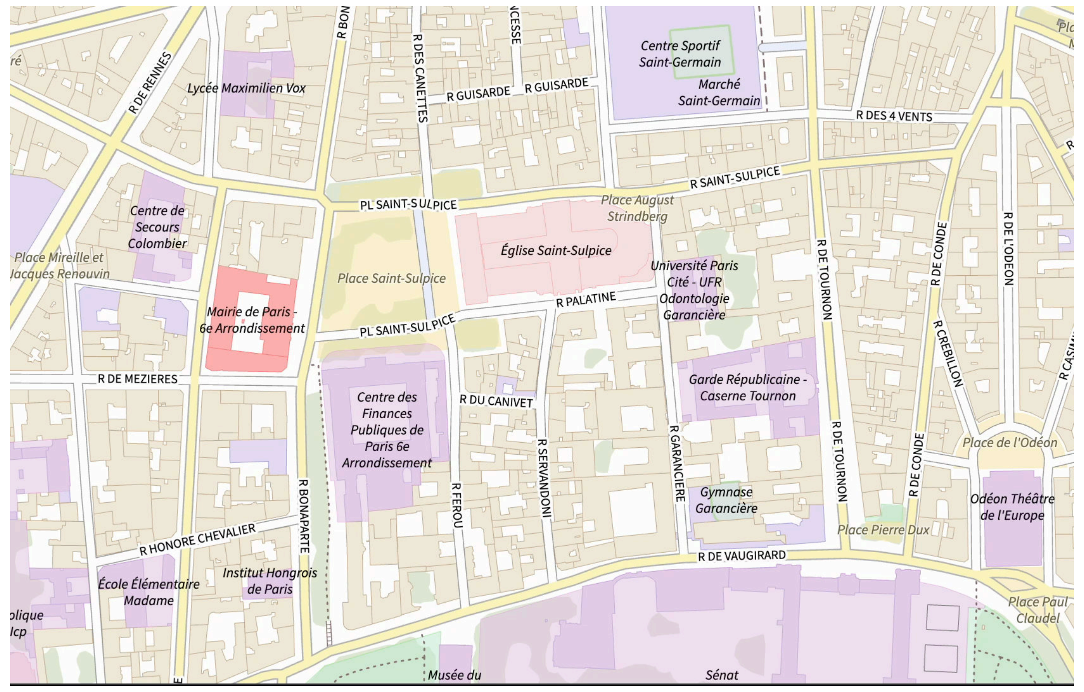
Coupled (FF)



Interaction technique: how the user controls the transition

Animated-from

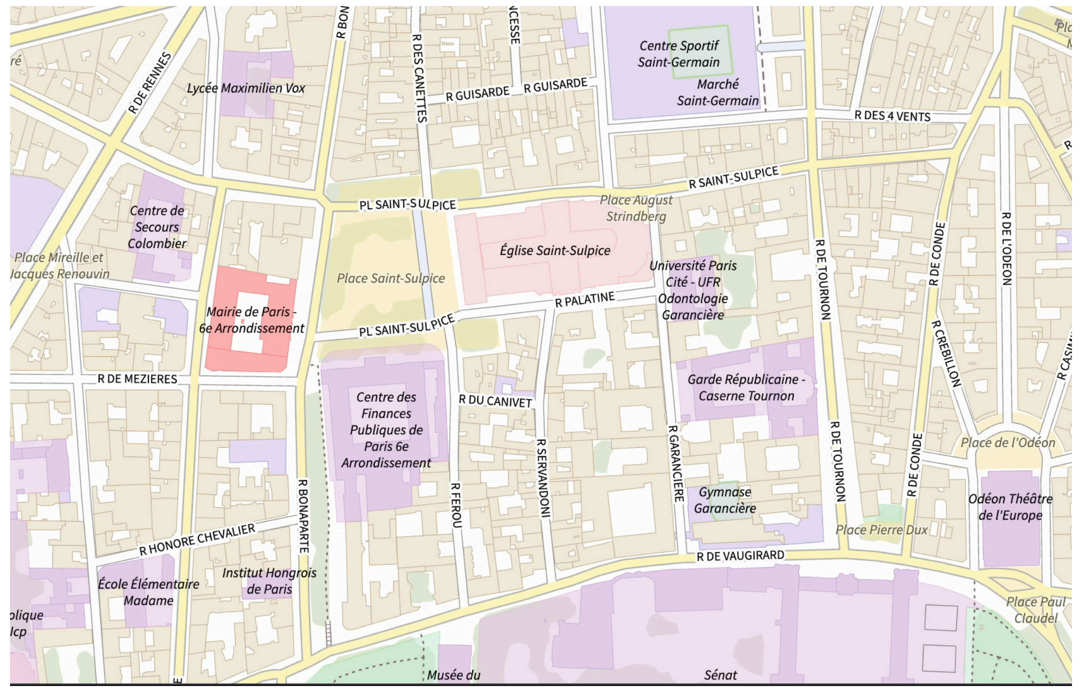
google like



from point

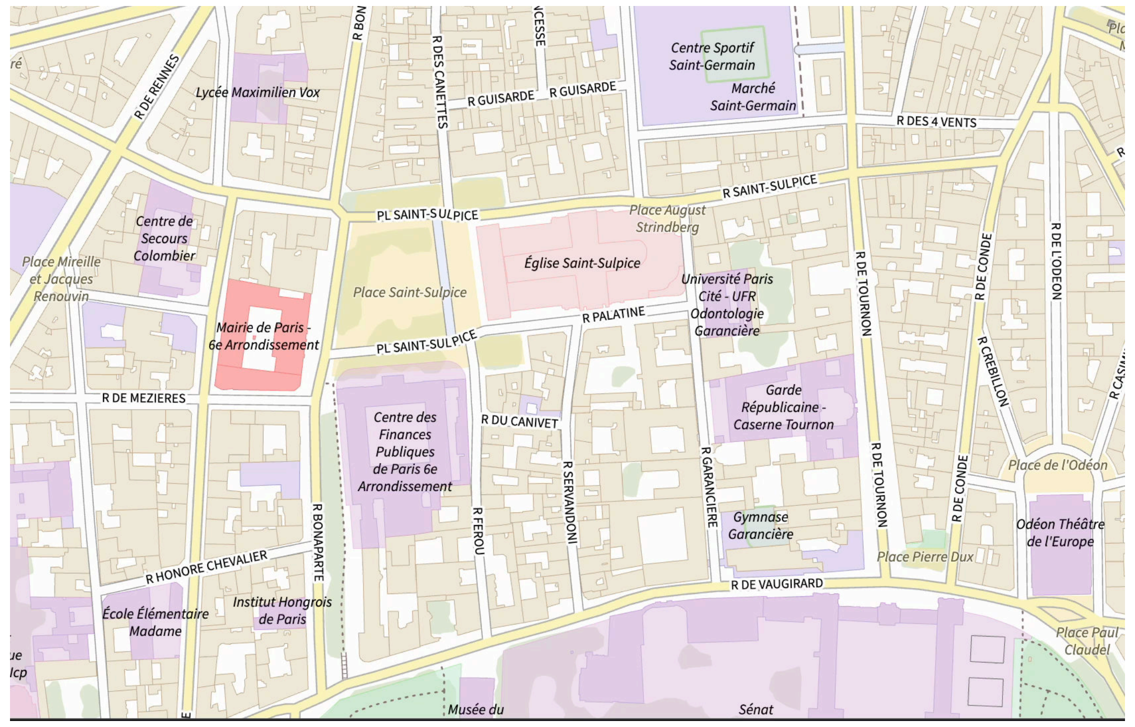
Animated-from-to

Danyluk et al, 2019



from and *at* point

Drag-along



from and *at* point progress

Less — User input — More

Interaction technique: how the user controls the transition

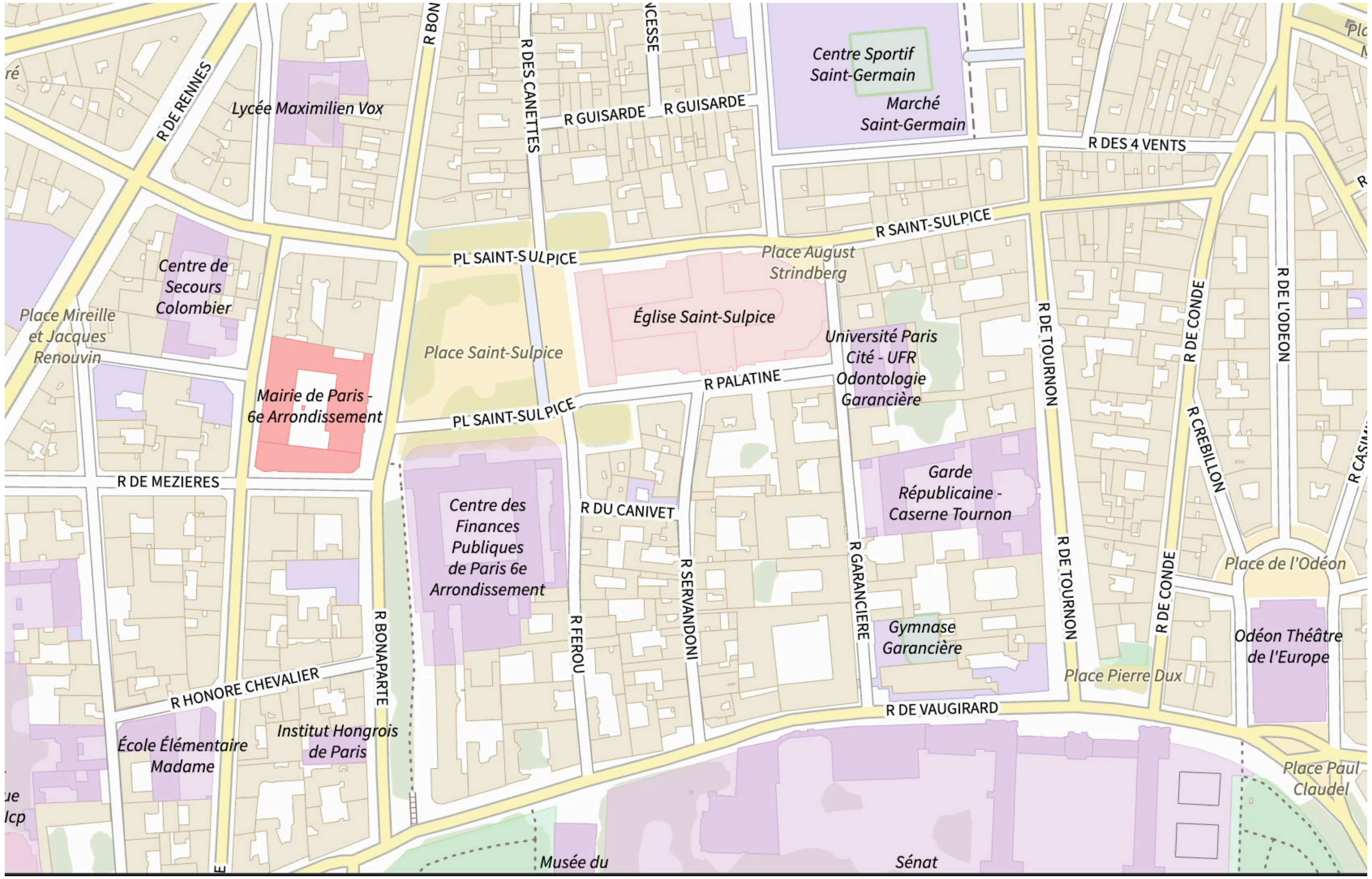
Animated-from

Interaction technique: how the user controls the transition

Animated-from-to

Interaction technique: how the user controls the transition

Drag-along



Exploratory study

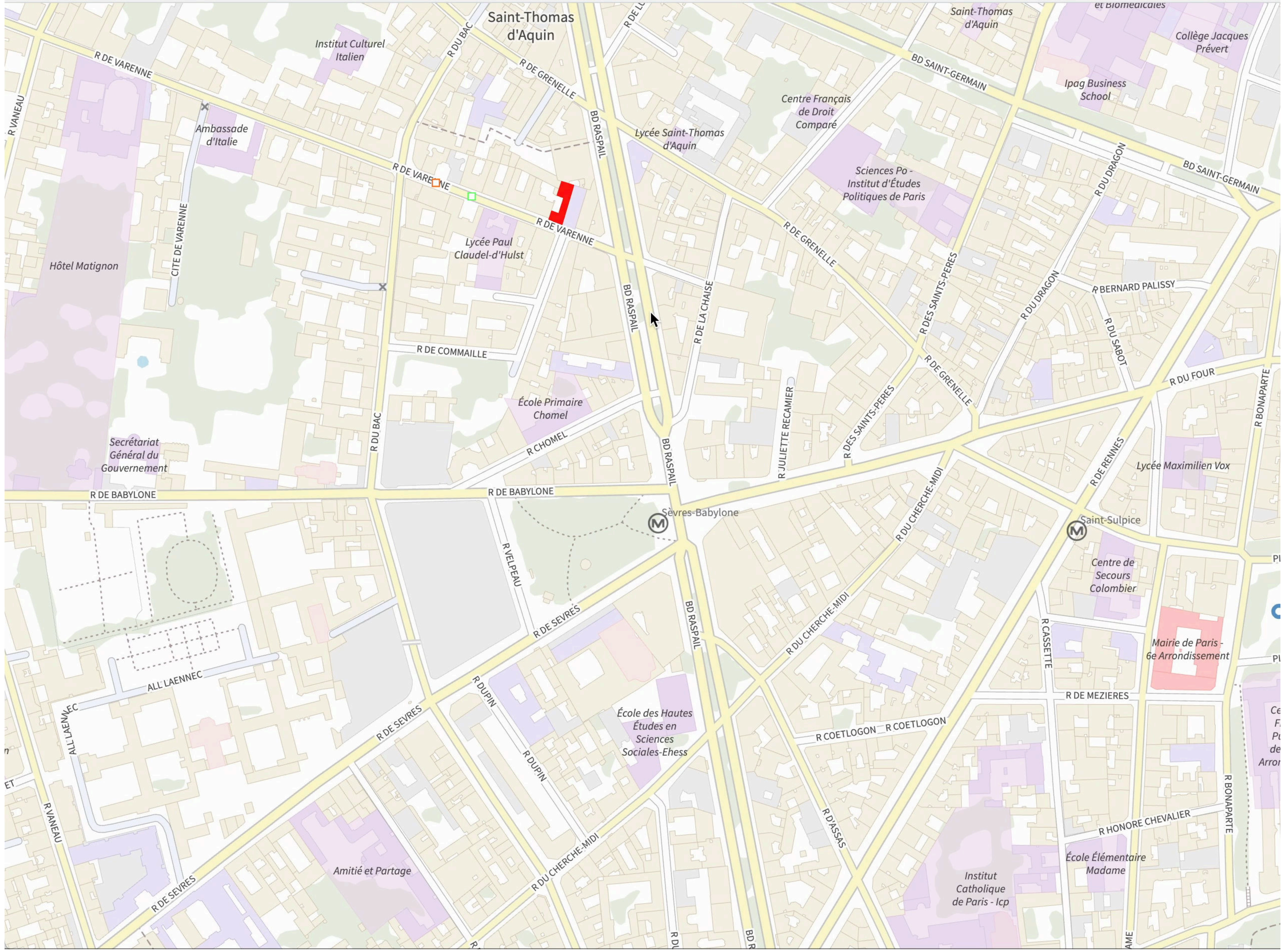
•**2 *camera transition*** [***Rotation First , Zoom First***]

X **2 *transition order*** [***Camera First , Features First, Coupled***]

X **2 *techniques*** [***Animated-from-to , Drag-along***]

- **24** participants
- **48** maps picked randomly in Paris, Lyon and Toulouse

Exploratory study



Results

- ***Animated-from-to*** was faster than ***Drag-along*** and more accurate
- ***Coupled*** and ***Features first*** were more accurate than ***Camera First***
- No difference between camera transitions

Discussion

- Participants relied on spatial structure
  - Specially using drag-along
  - Participants’ working regularly with geographical data
  - Relied on the overview to link the 2D and the 3D representation

Comparing to the baseline:

we propose to compare **7** combinations of ***transitions*** and ***interaction techniques*** based on the first experiment results and test different **animation durations**

User study II

- **7 techniques:**
  - ***Animated-from-to*** with 0s animation, 2s animation and 6s animation
    - Using ***Rotation First*** and ***Features First***
  - ***Animated-from*** with 0s animation, 2s animation and 6s animation
    - Using ***Rotation First*** and ***Features First***
  - ***Drag-along*** with ***Zoom First*** and ***Coupled***

- **21** participants (not the same than before)
- **49** maps picked randomly in Paris, Lyon and Toulouse

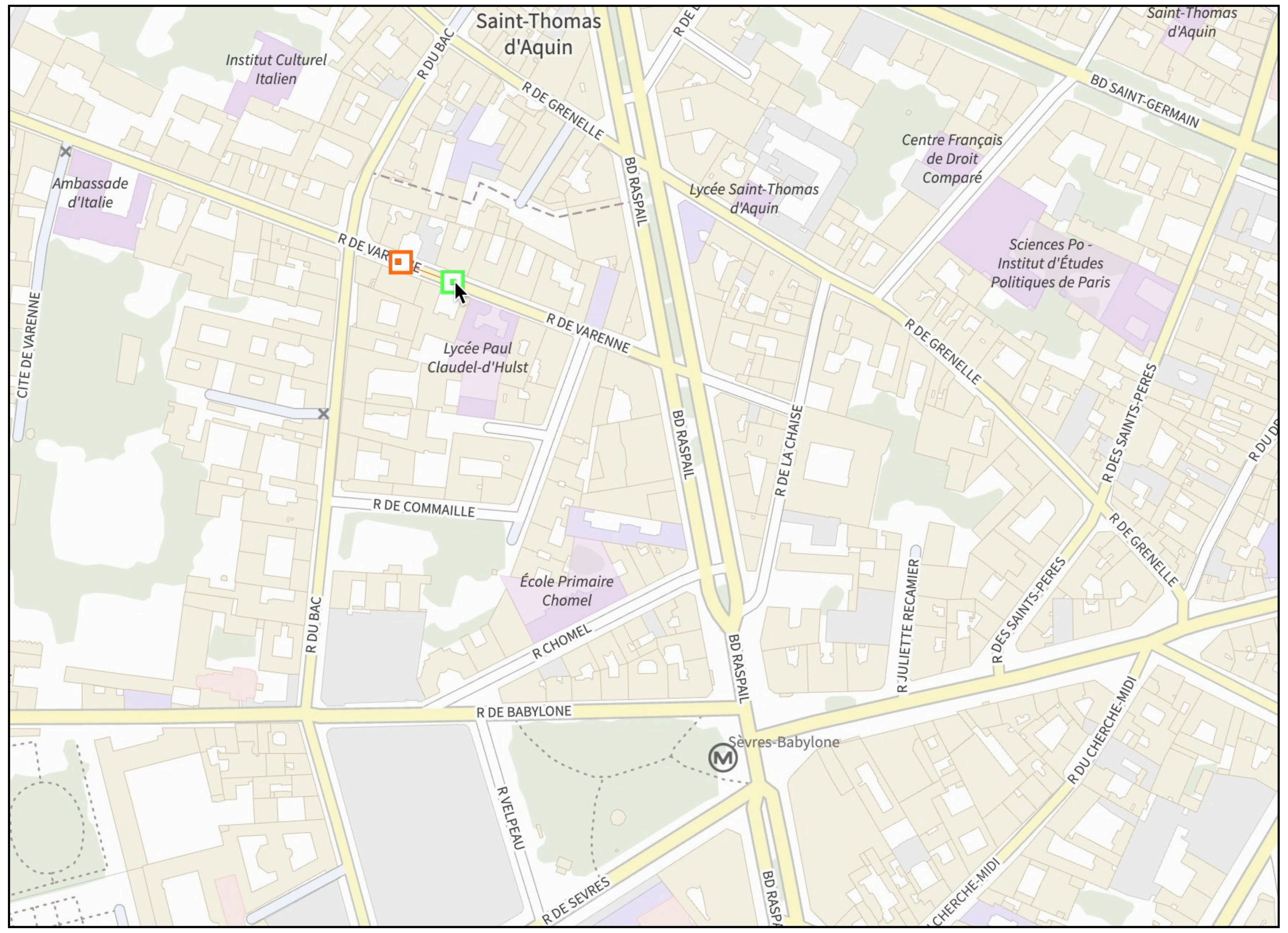
Results

- Our results suggest that:
  - Having an **animated transition** between the two views is beneficial
  - **Longer** transitions are better
  - More input **before** the transition seems to be better

- Limitations
  - Tested a subset of the design space
  - Environment not realistic

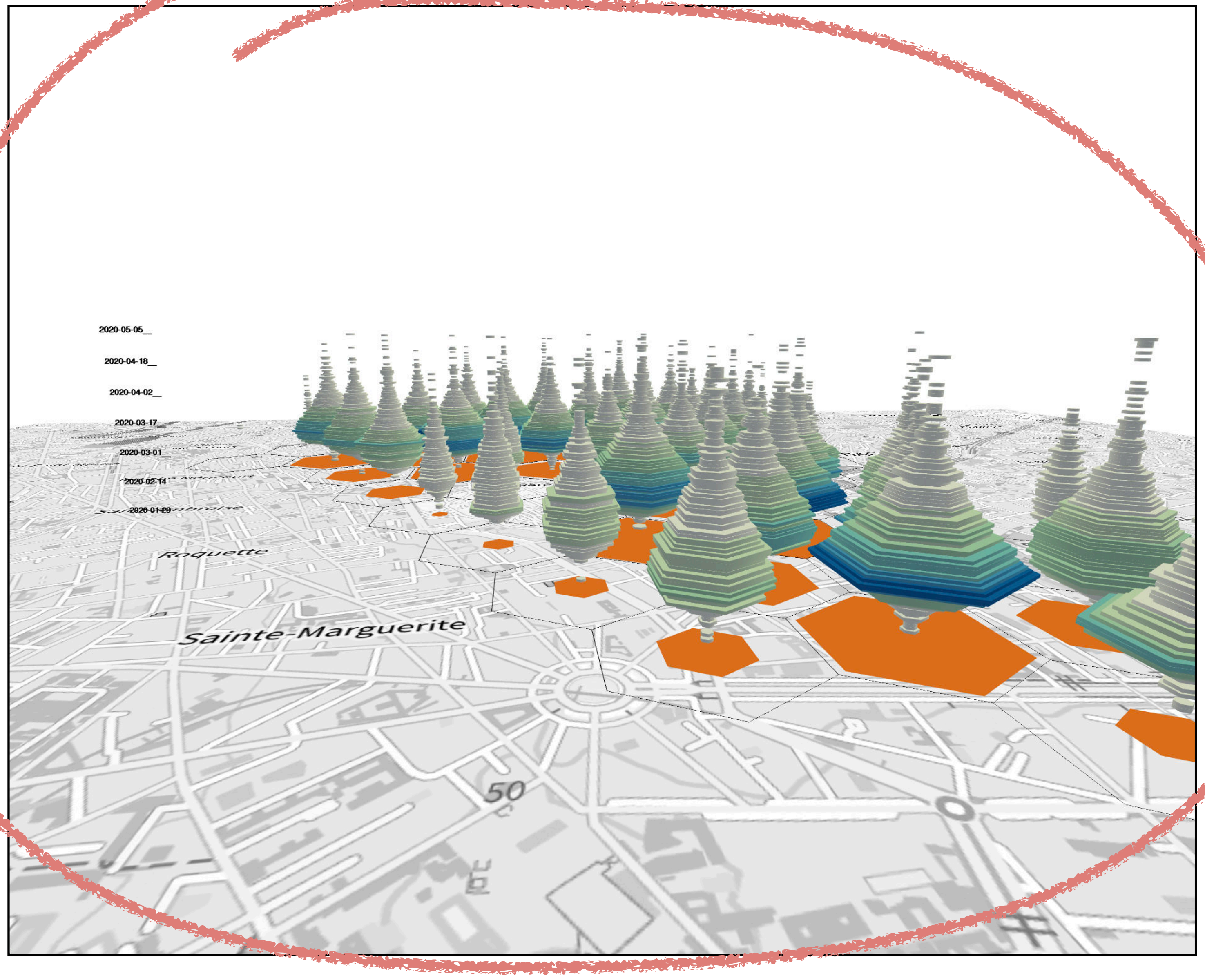
[Maria-Jesus Lobo, Mathieu Brédif, Sidonie Christophe, Work in progress]

Map views



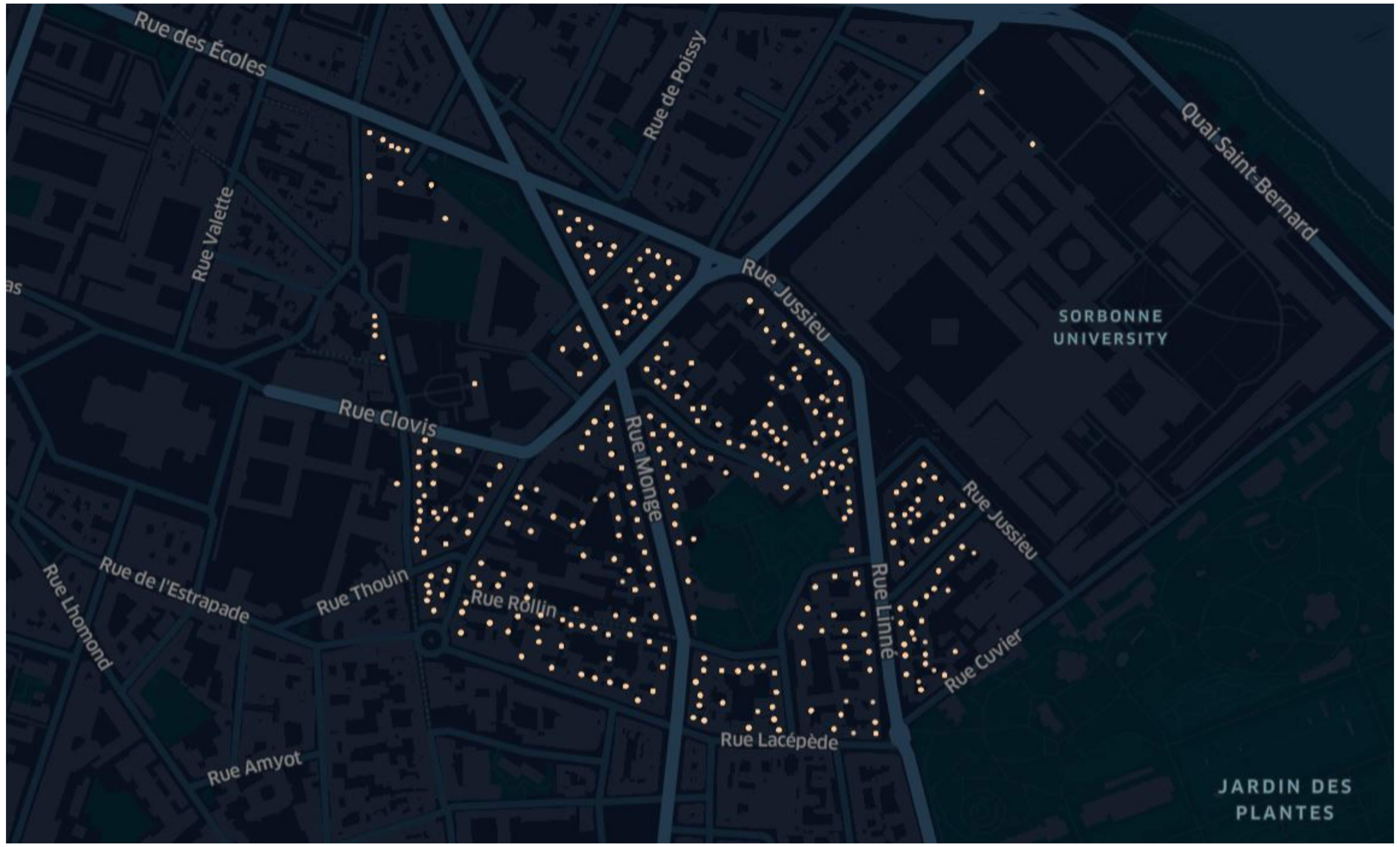
2D top-down to 3D immersive transitions

Spatio-temporal visualisations



Géopidemio

Simulated Covid data visualisation



Data from M. Colomb - ICI Project

How to visualise COVID events to identify

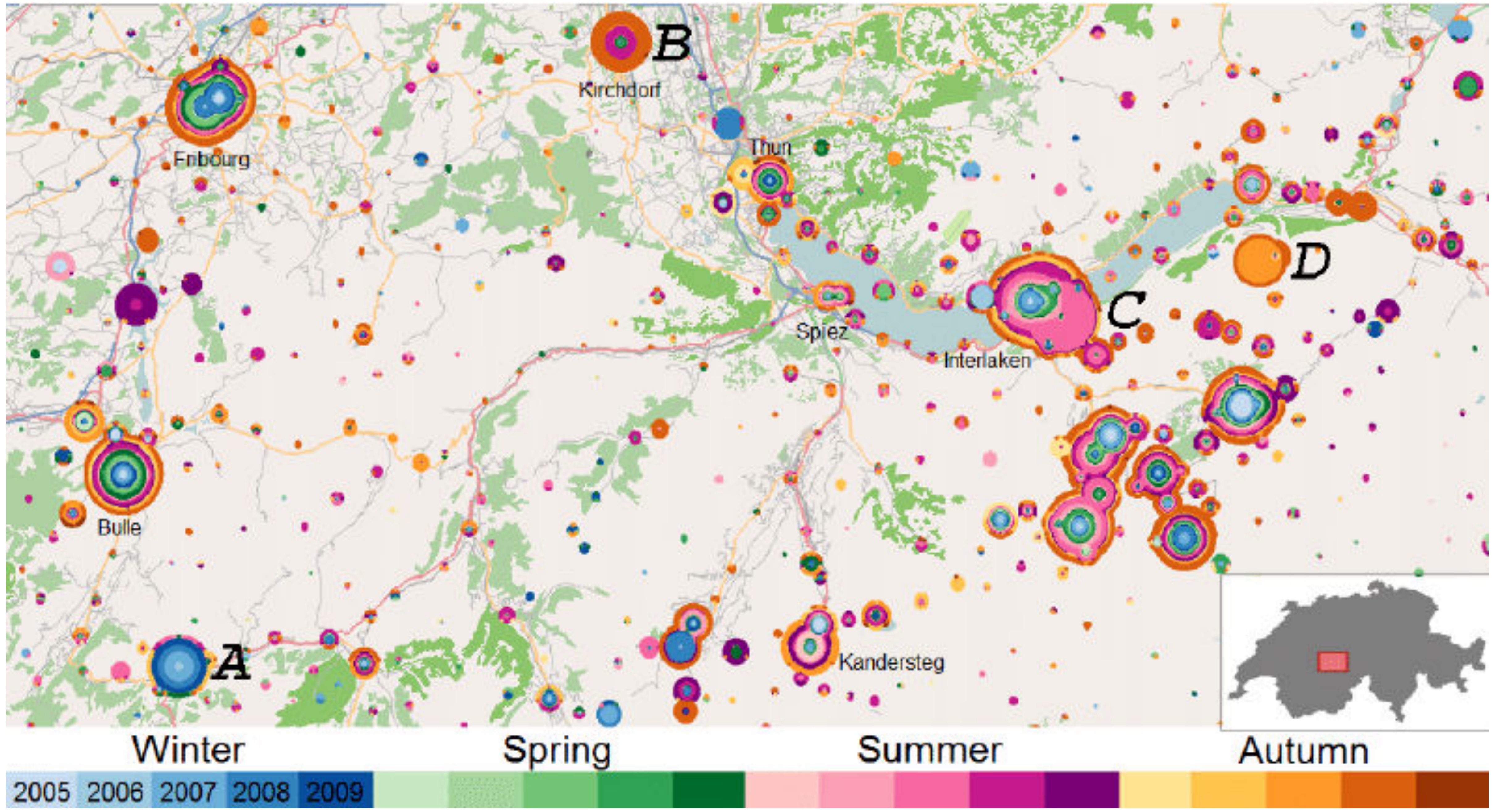
… epidemic phases

… spatio-temporal clusters

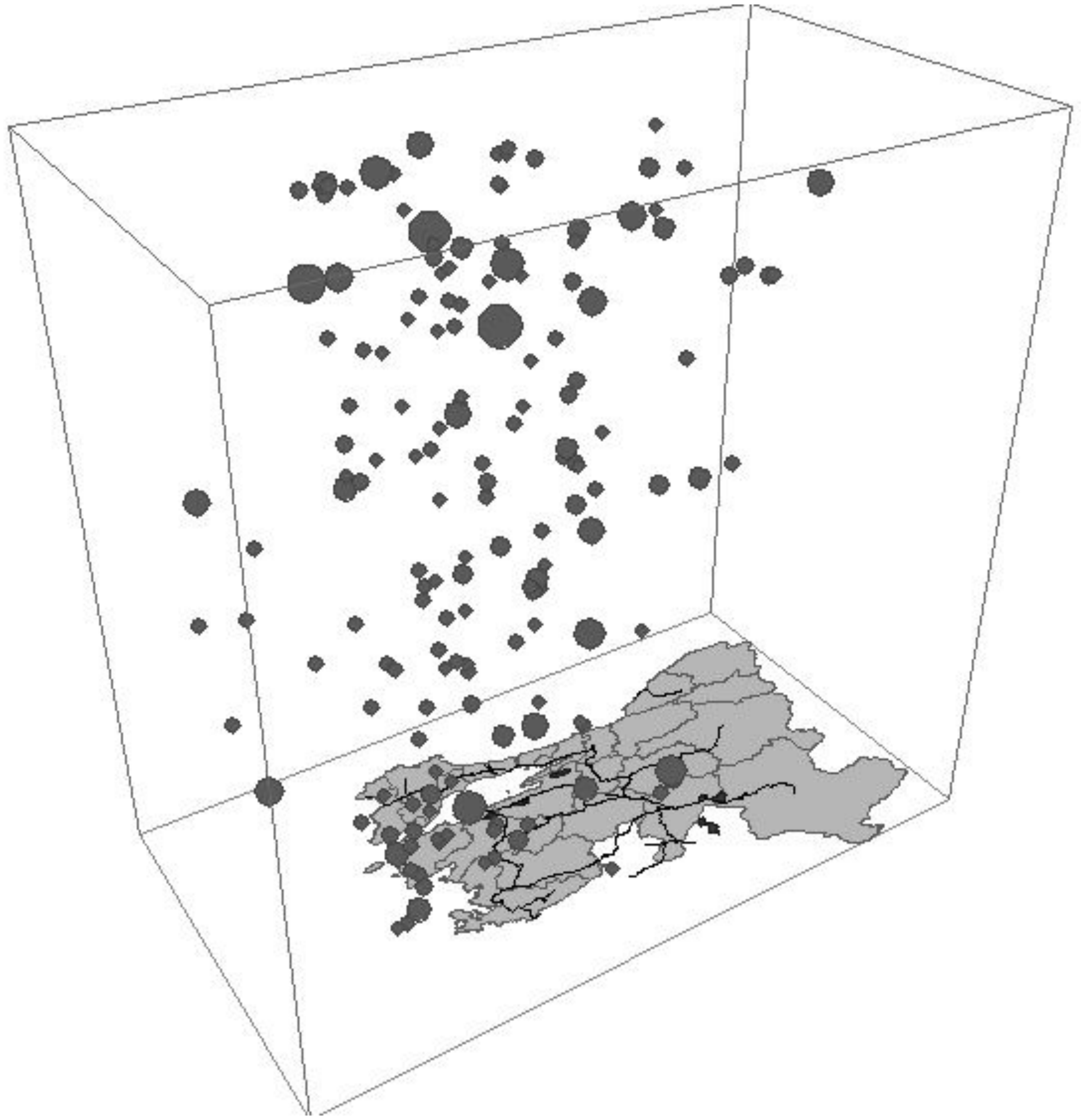
… propagation axes

Gautier, J., Lobo, M. J., Fau, B., Drugeon, A., Christophe, S., & Touya, G. (2021, December). COVID-19 geoviz for spatio-temporal structures detection. In *30th International Cartographic Conference (ICC 2021)* (Vol. 4, p. 37). Copernicus Publications.

Existing approaches



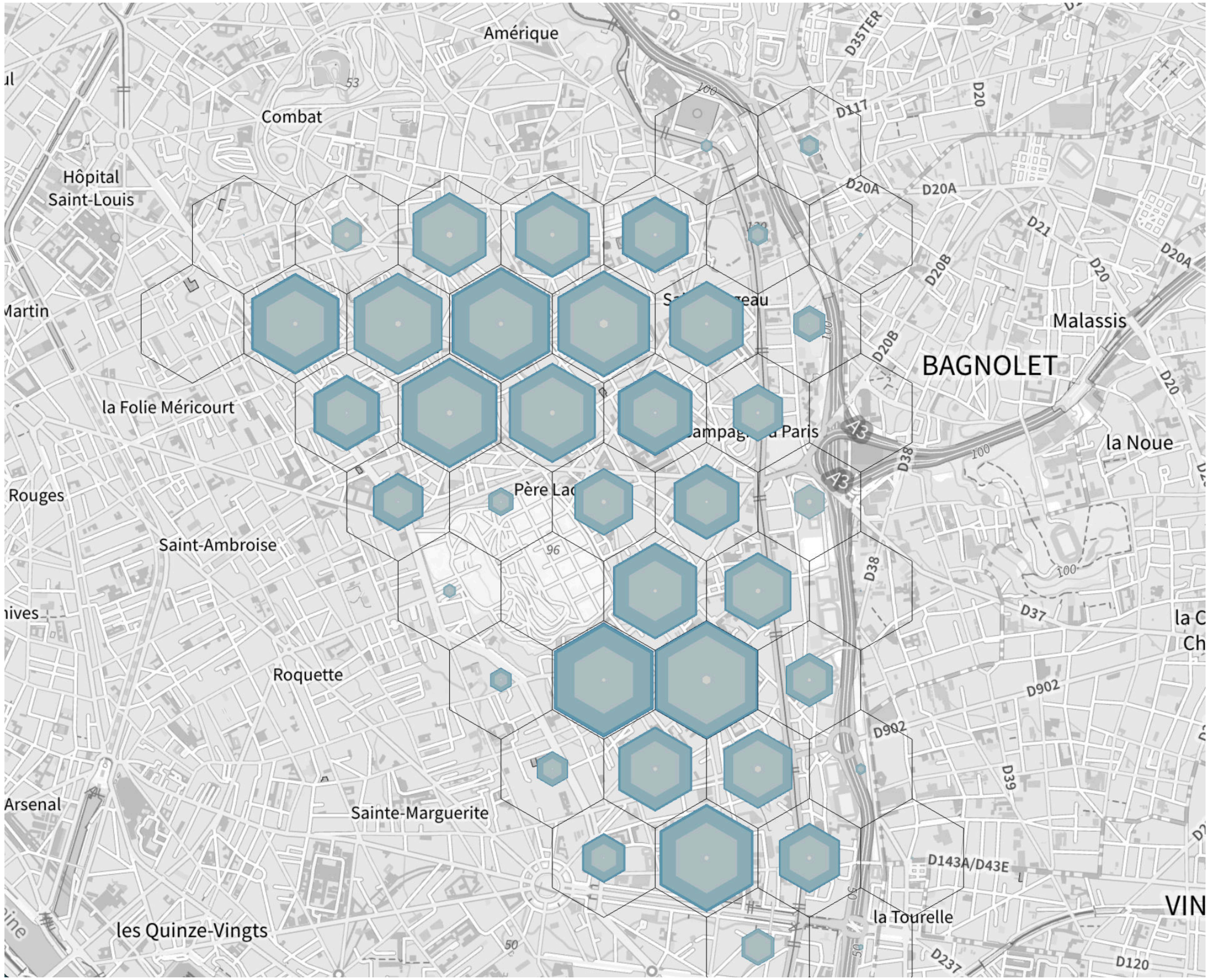
[Bak et al., 2009]



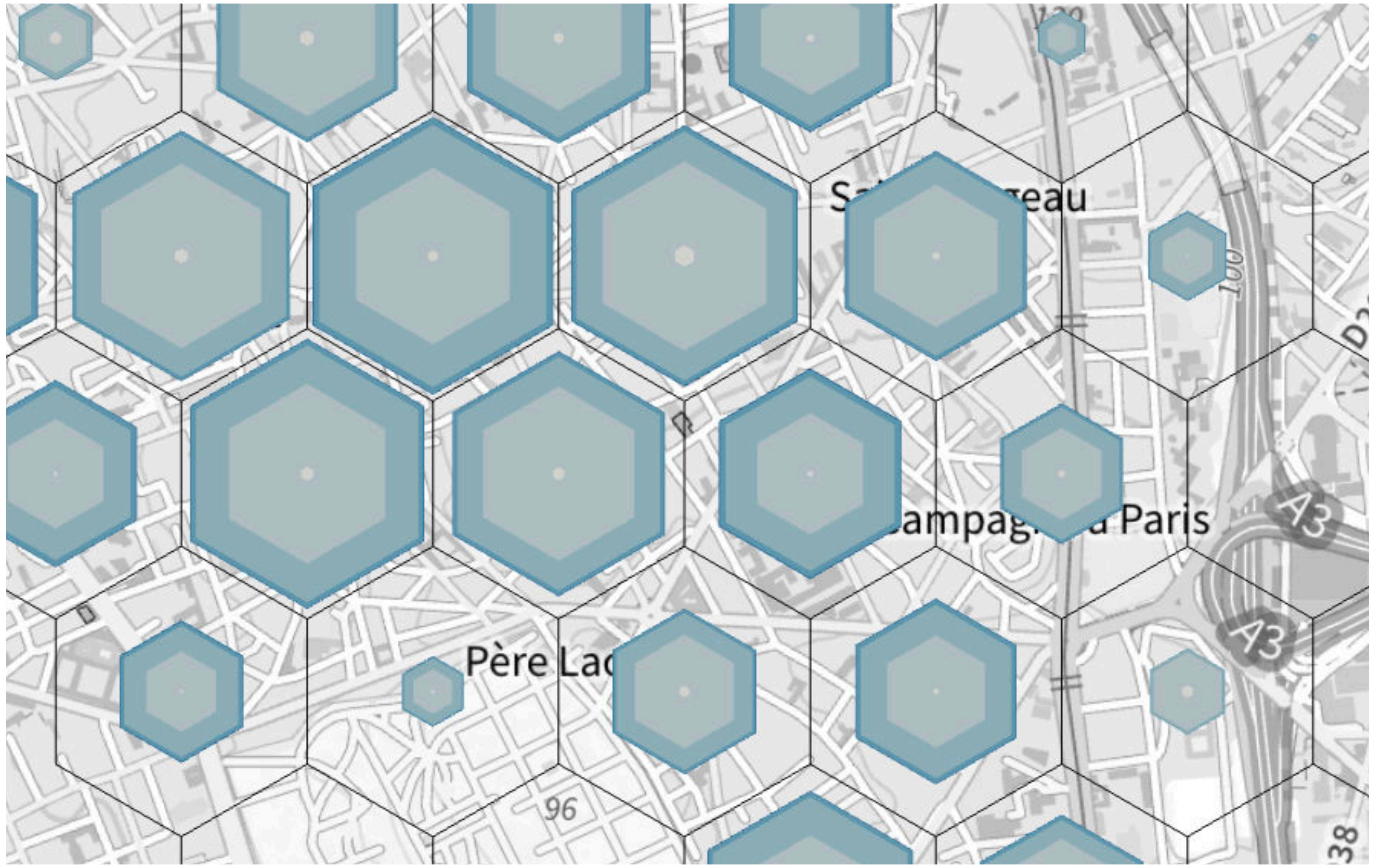
[Gatalsky, Peter et al., 2004]

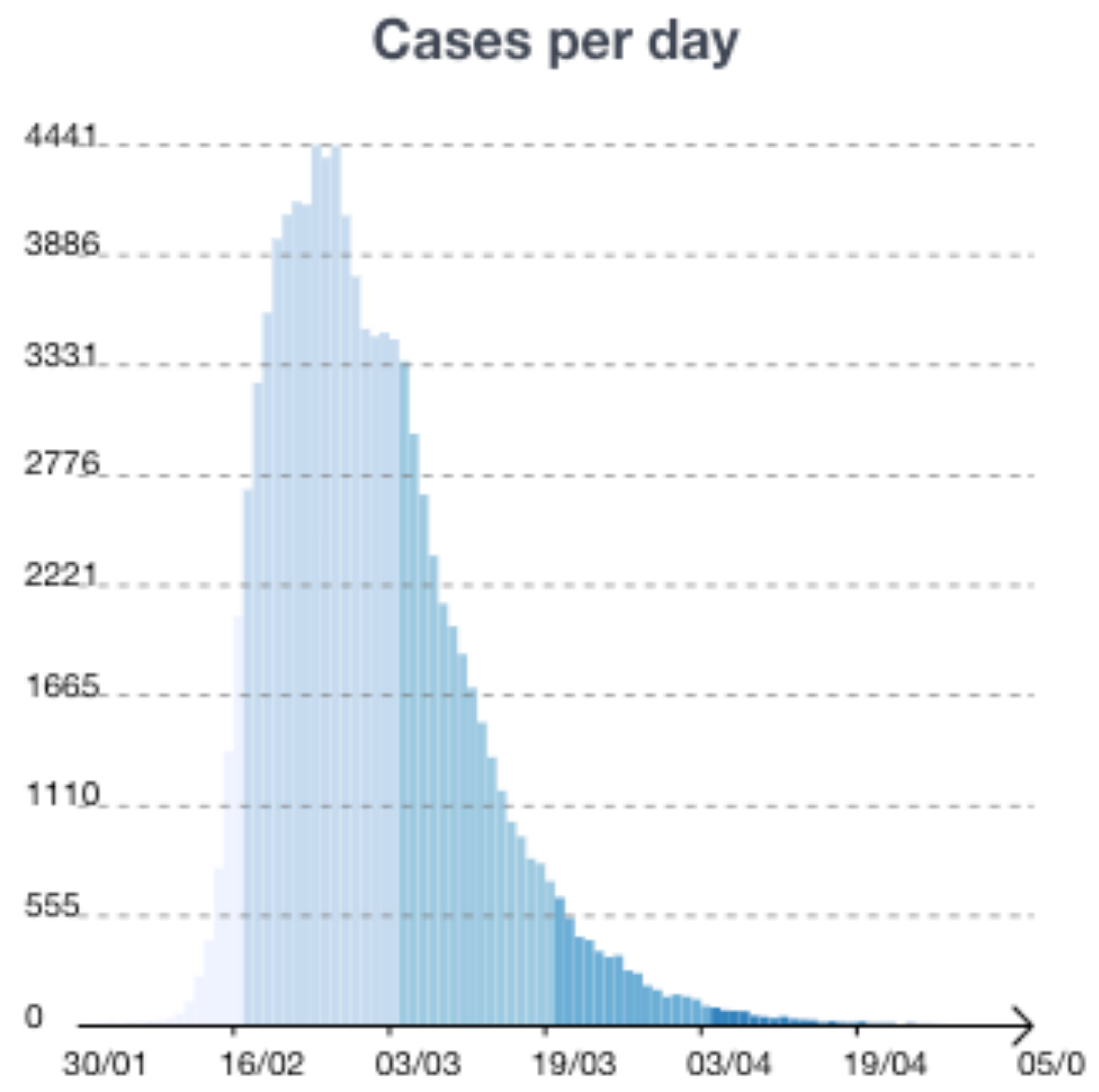
Can we **combine** these two approaches through **interactive transitions**?

2D View: based on the growth ring map

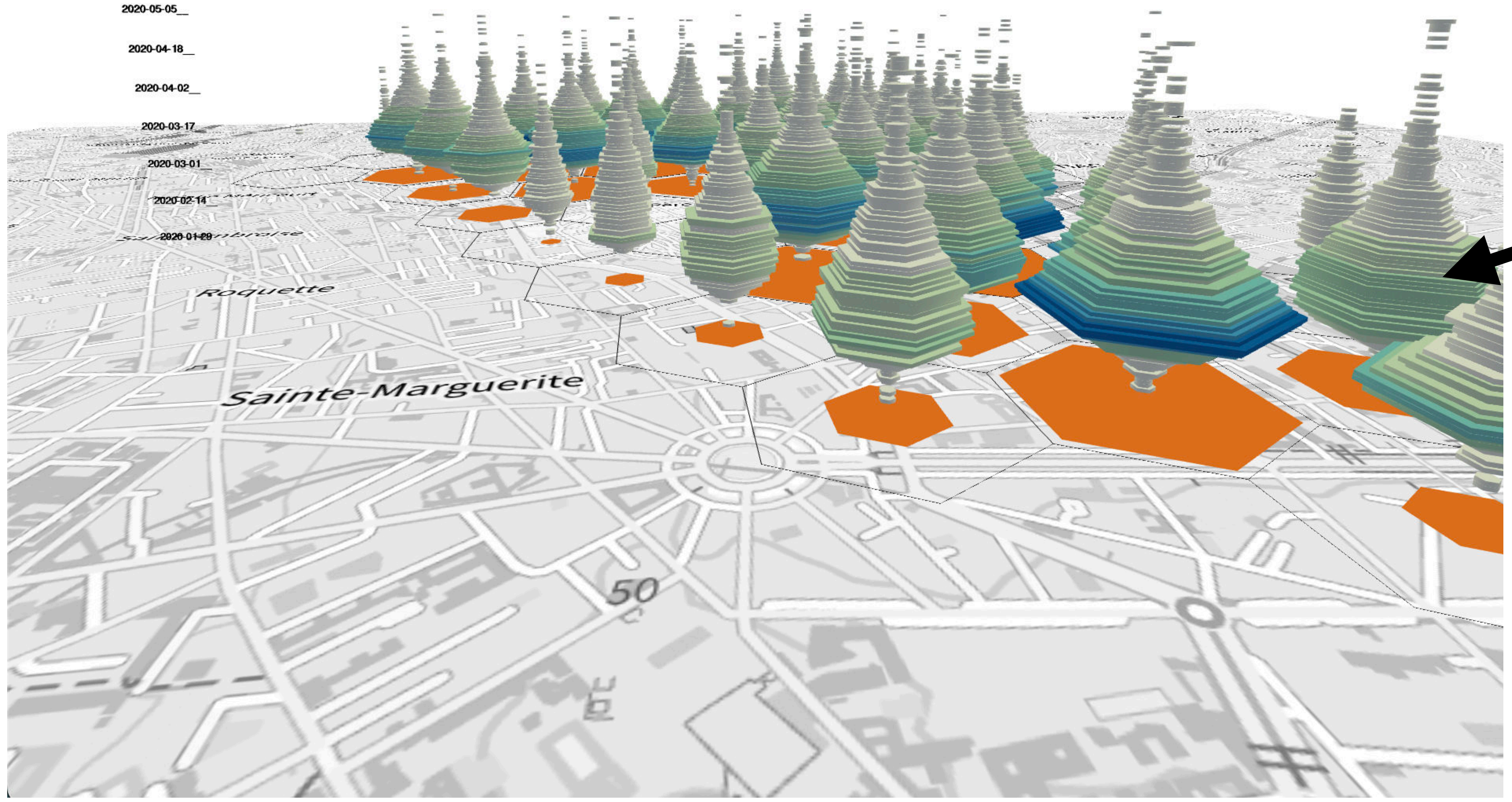


2D View: based on the growth ring map



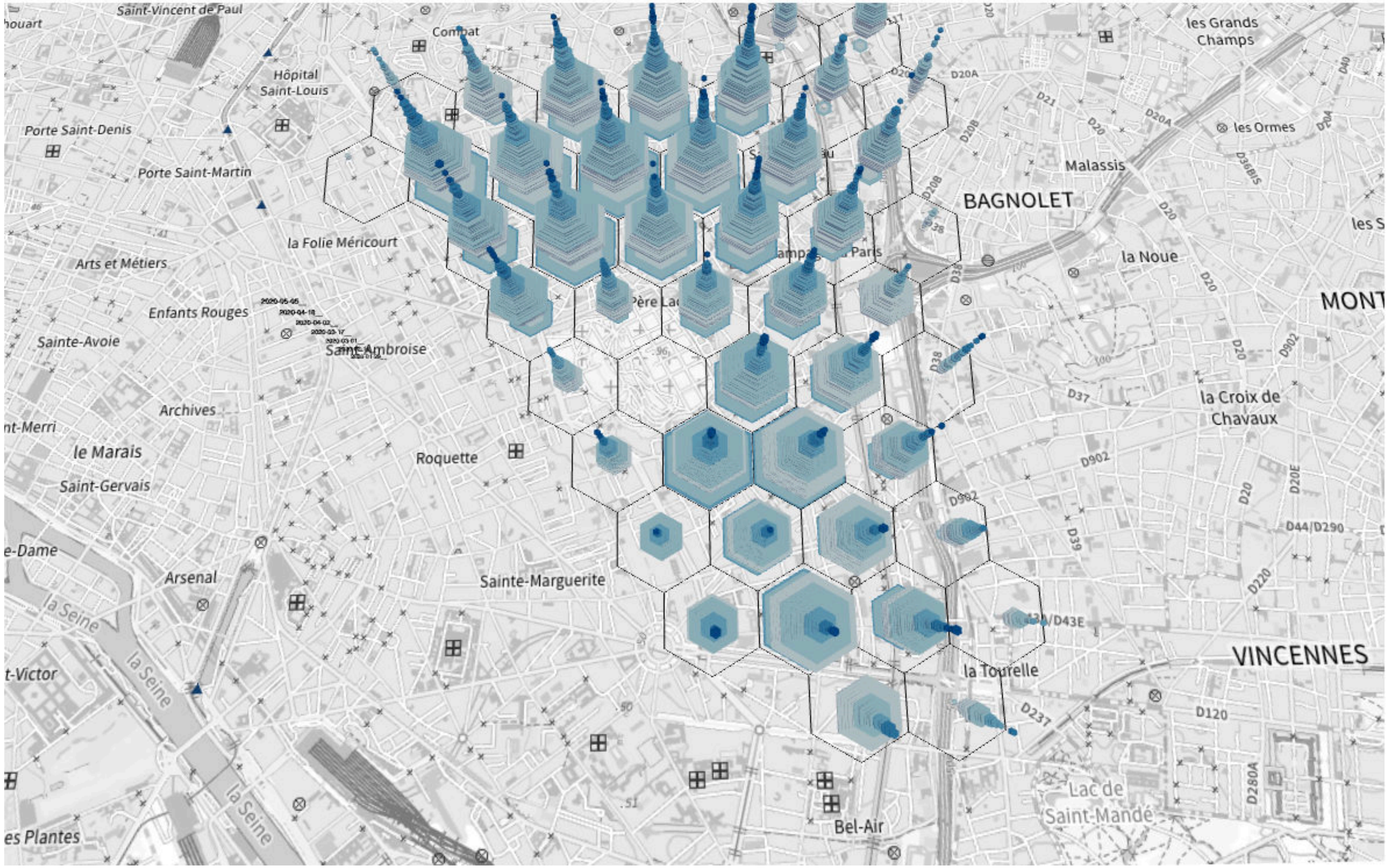


3D View: Spatio-temporal Cube

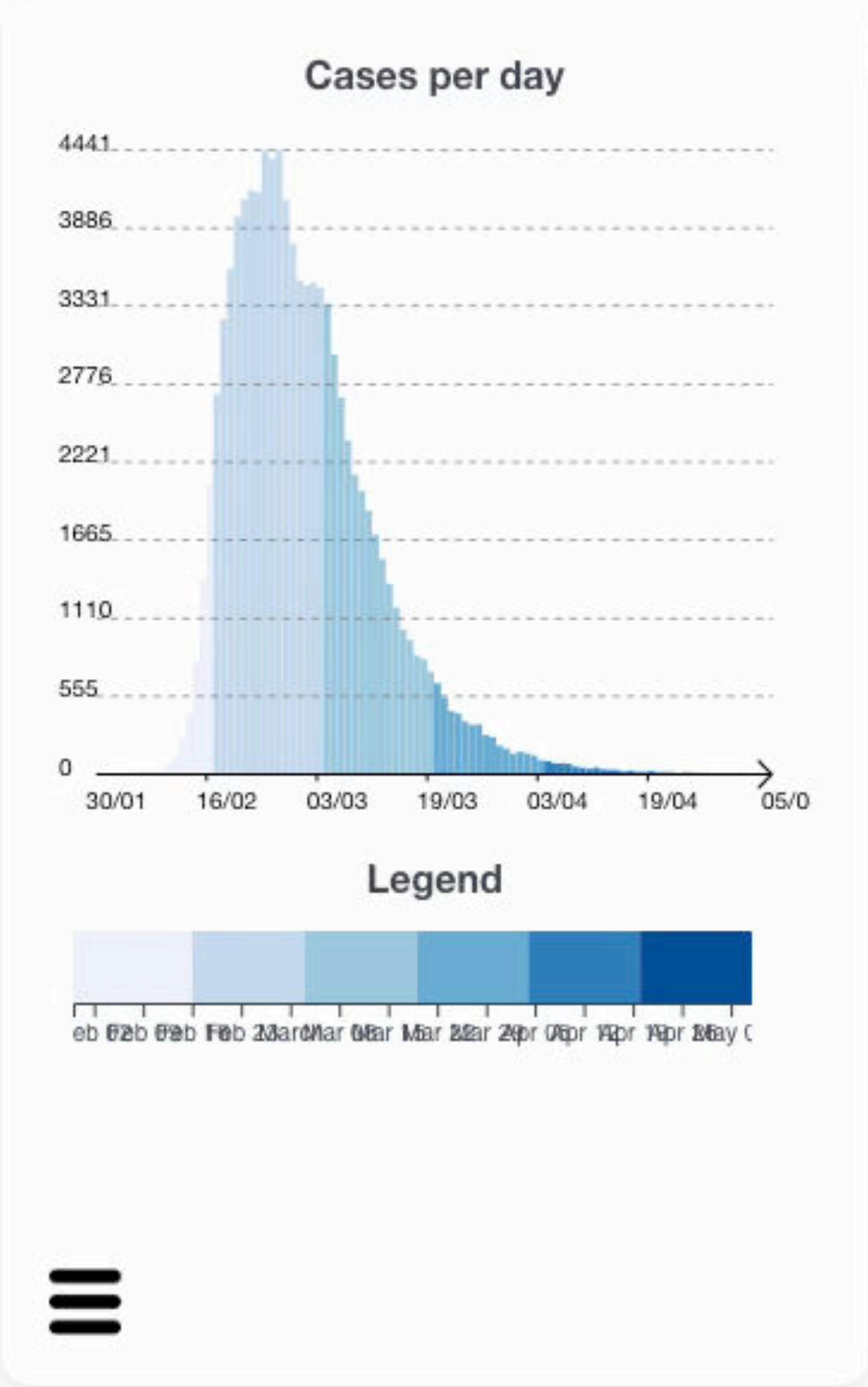


An hexagon represents a day

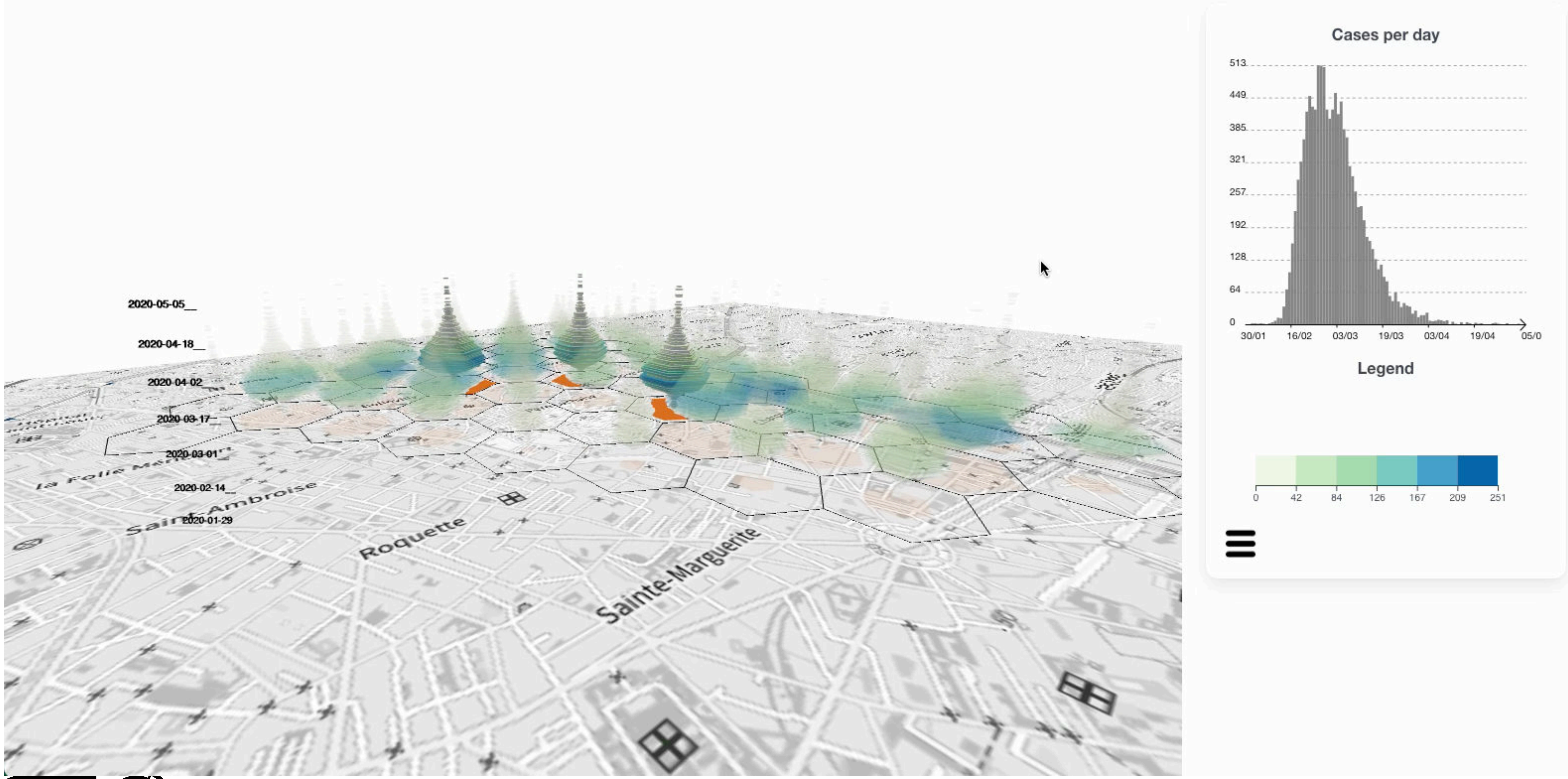
Transition view



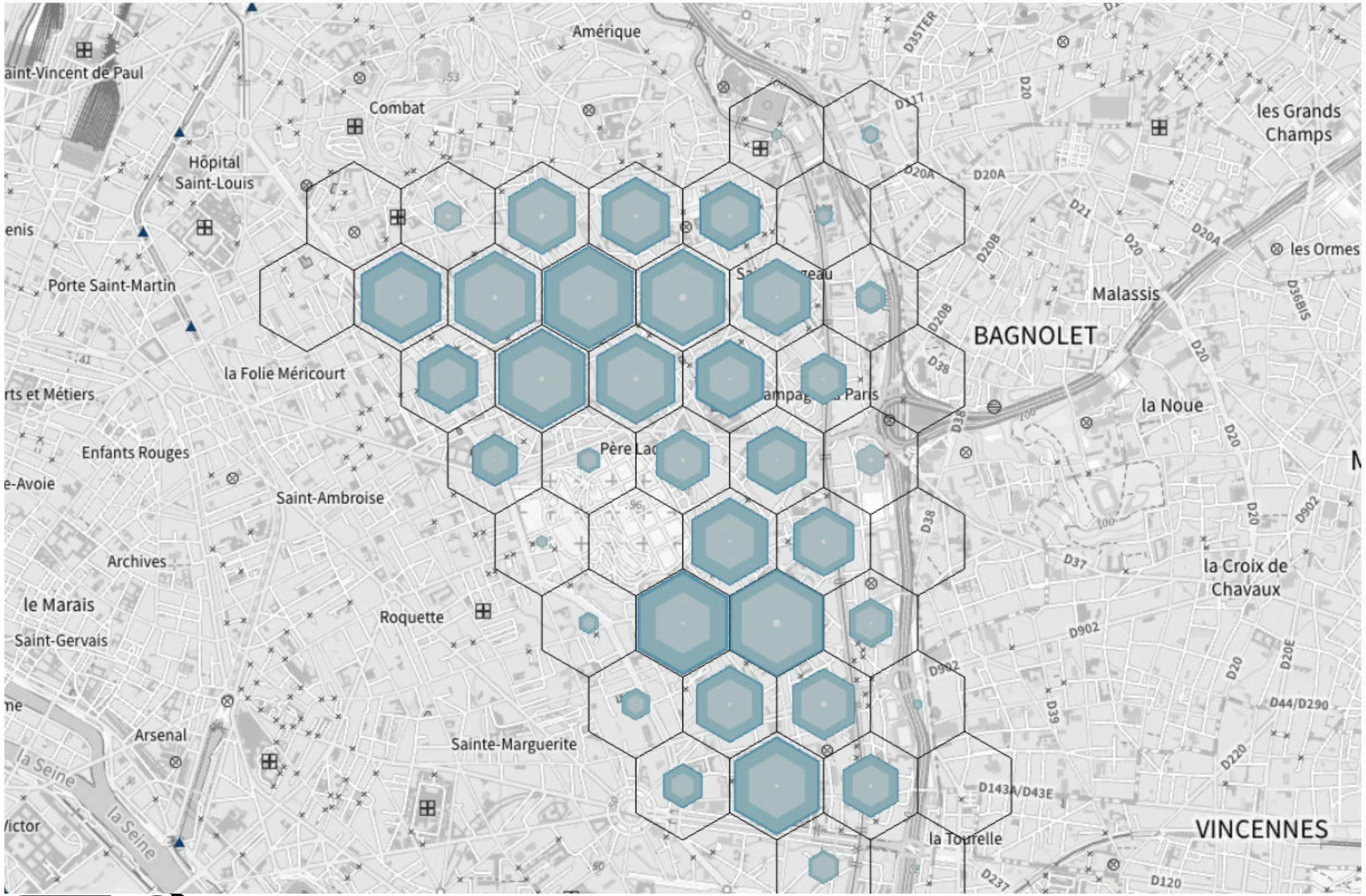
Transition view

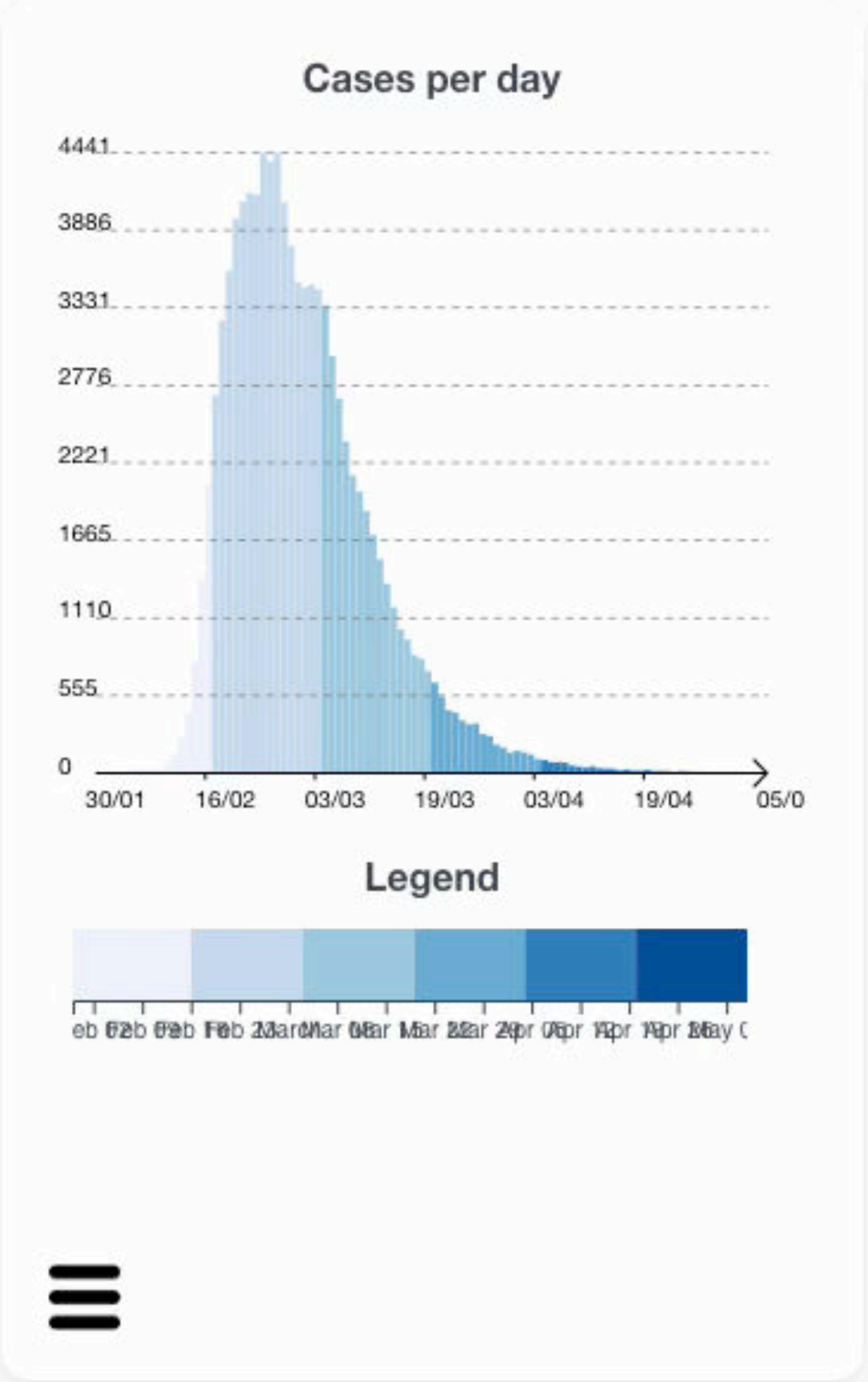


Region selection



Day selection

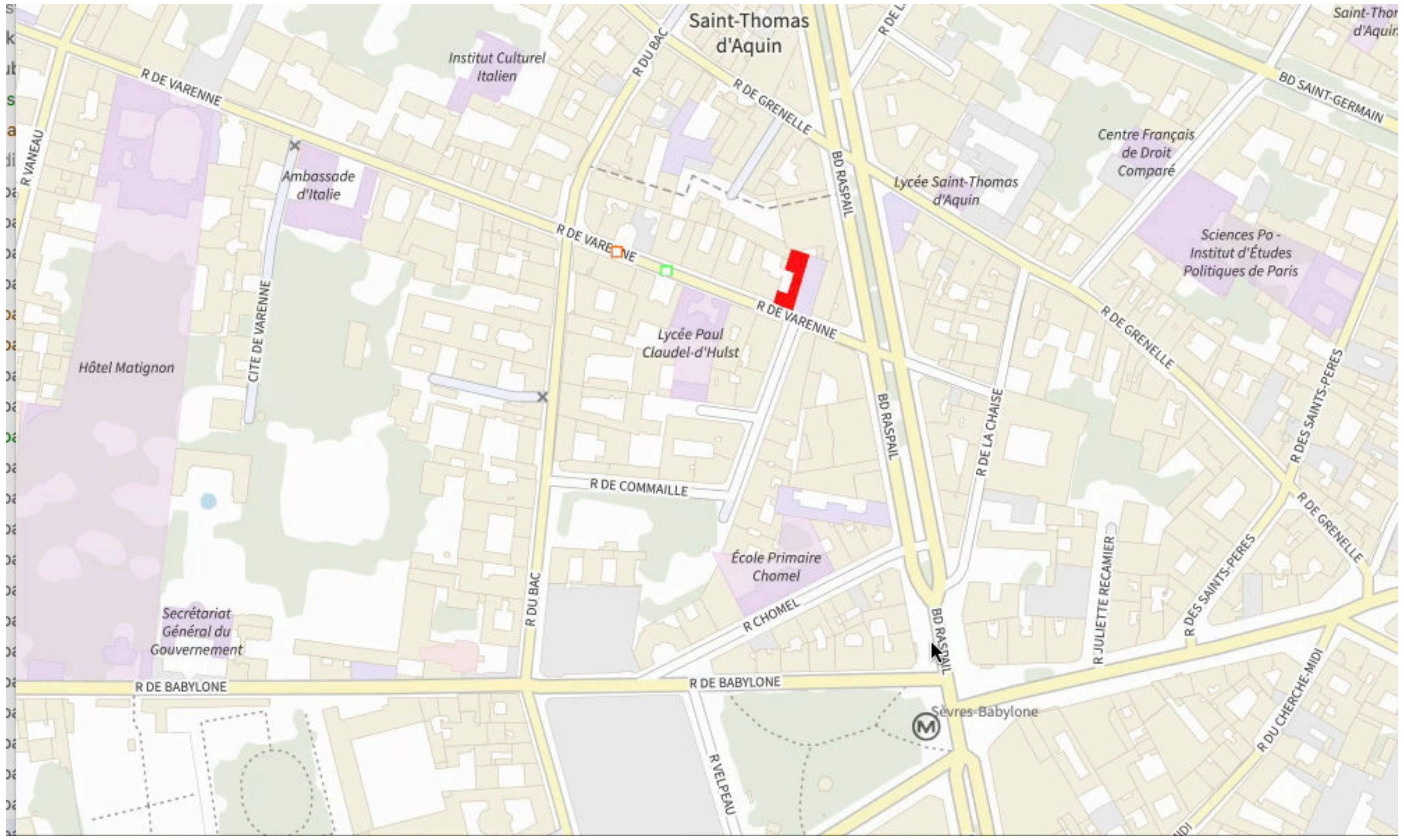




Qualitative feedback

- Positive feedback in general
- But many improvement suggestions:
  - Choose hexagonal grid in a smarter way
  - Option to compare multiple scenarios
  - Add point of interest such as schools, hospitals
- Another evaluation is coming soon

Future work


Saint-Thomas d'Aquin
Institut Culturel Italien
R DE VARENNE
R DE GRENELLE
BD RASPAIL
Lycée Saint-Thomas d'Aquin
Centre Français de Droit Comparé
BD SAINT-GERMAIN
Ambassade d'Italie
Sciences Po - Institut d'Études Politiques de Paris
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Hôtel Matignon
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R VANEAU
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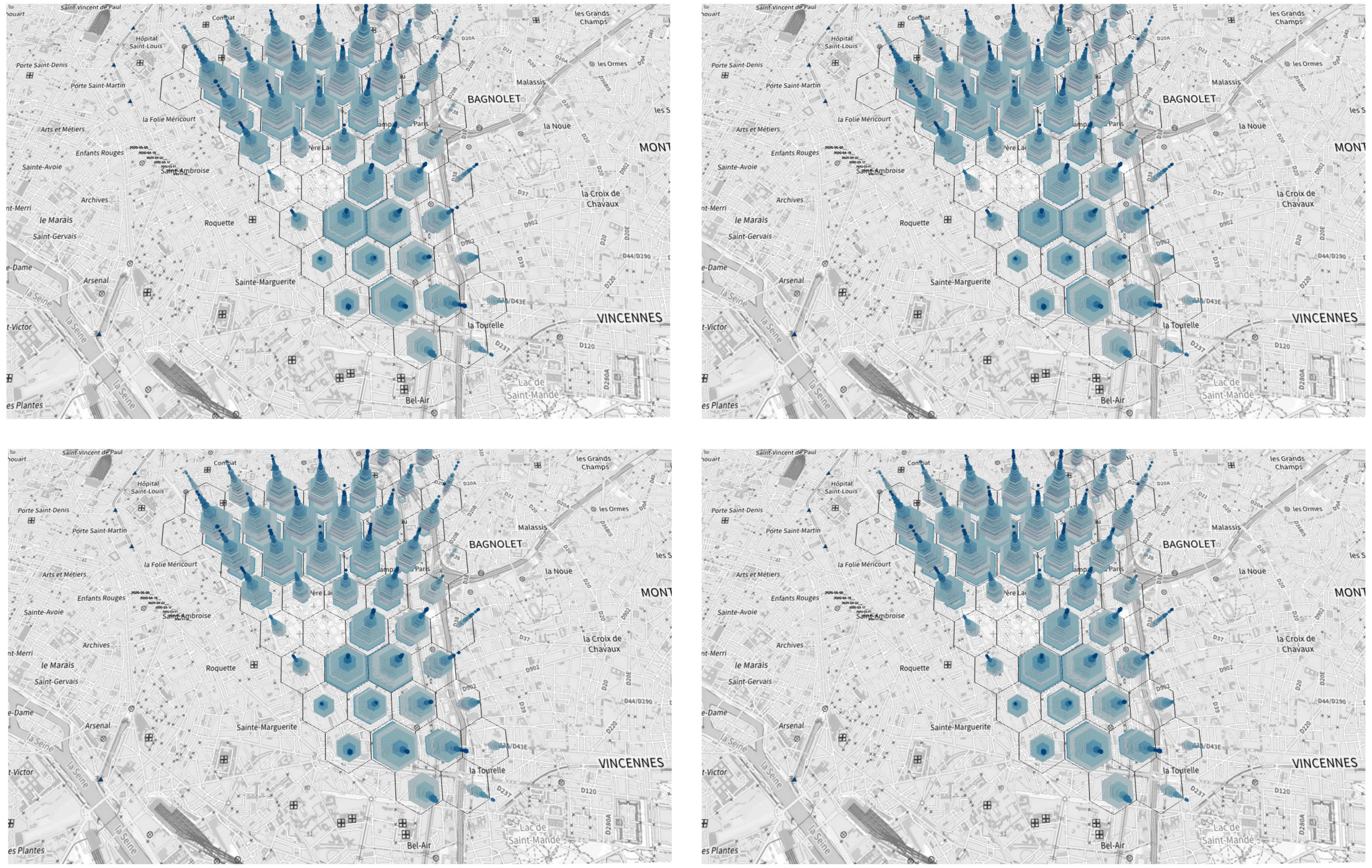
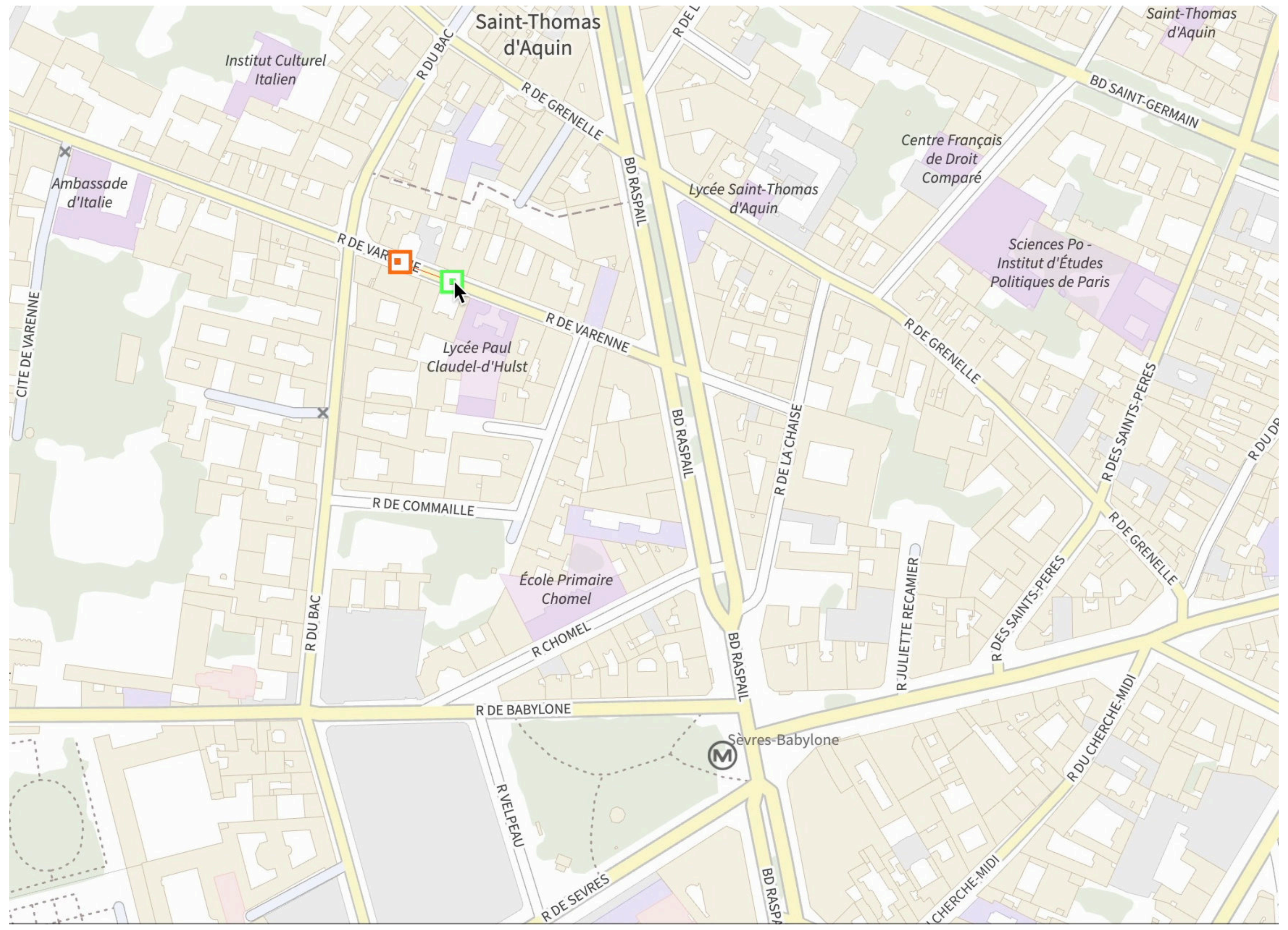
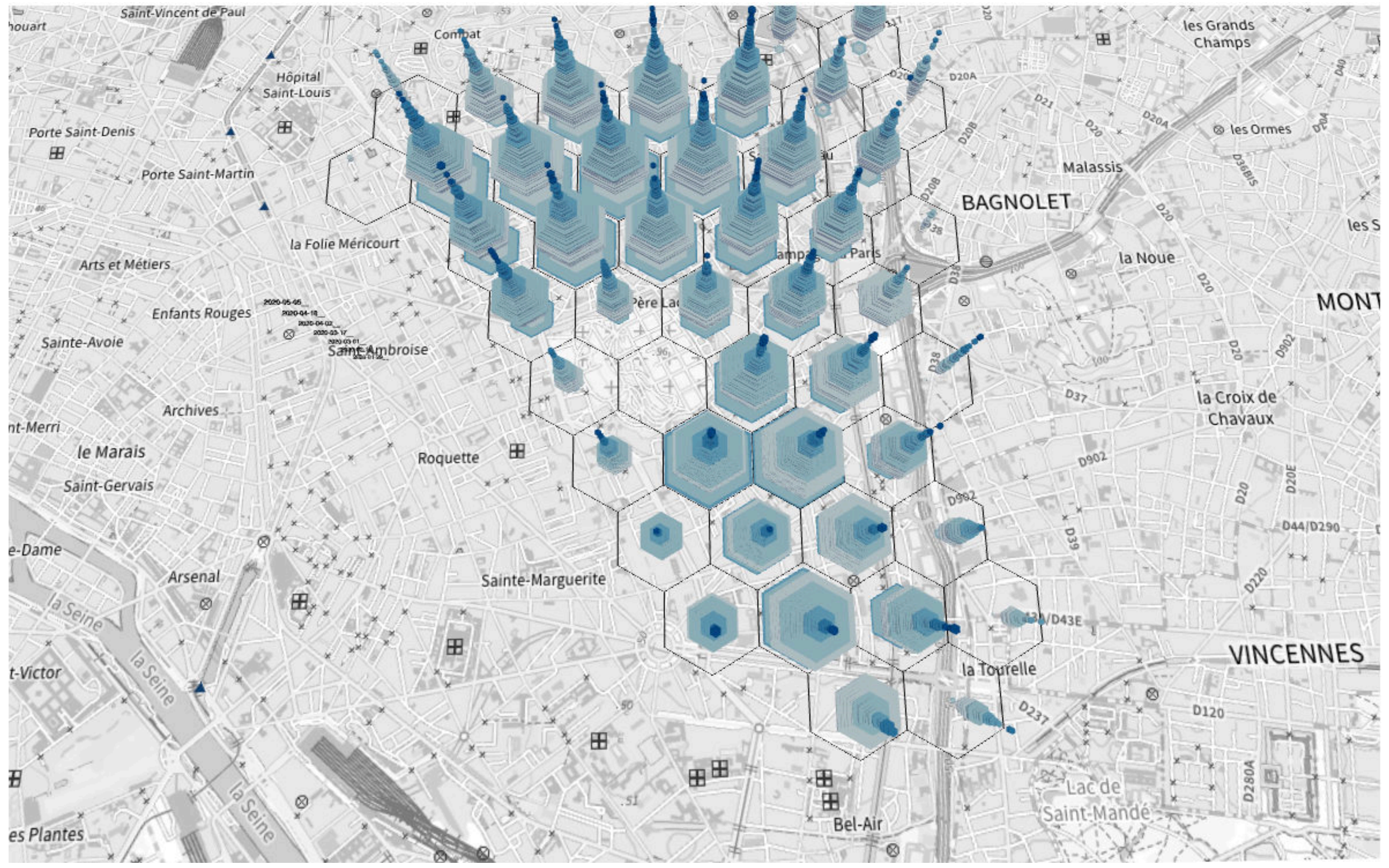
R DES SAINTS-PERES
R DE COMMAILLE
École Primaire Chomel
R CHOMEL
R DU BAC
R JULIETTE RECAMIER
Secrétariat Général du Gouvernement
R DE BABYLONE
Sèvres-Babylone
R VELPEAU
R DU CHERCHE-MIDI


BAGNOLET
VINCENNES

Transitions entre vues 2D et 3D pour des visualisation des données spatio-temporelles





María Jesús Lobo, Jacques Gautier, Mathieu Bredif, Sidonie Christophe
https://www.umr-lastig.fr/geovis/